

Distortive Effects of Deposit Insurance: Administrative Evidence from Deposit and Loan Accounts ^{*}

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August 6, 2026

Abstract

We analyze how increasing deposit insurance coverage during crises shapes deposits and lending. We exploit a reform of Denmark's deposit insurance during the Global Financial Crisis and administrative datasets, including the universe of deposits. Increasing insurance coverage prevents weaker banks from further losing large (previously uninsured) deposits during crises, enabling continued credit supply. However, increasing insurance facilitates reallocation of large deposits from stronger to weaker banks, and increases credit supply by weaker banks to ex-ante riskier firms with higher ex-post defaults. Consistently, banks that gained most deposits under unlimited insurance are themselves more likely to fail, particularly ex-ante weaker banks.

Keywords: Deposit Insurance, Deposit Allocation, Deposit Rates, Credit Allocation, Financial Crises

JEL Classification: G01, G21, G28, E51

^{*}We thank Jason Allen, Matthew Baron, Diana Bonfim, Kim Cramer, Eduardo Davila, Giuseppe Ferrero, Chacko George, Itay Goldstein, Florian Heider, Skander Van den Heuvel, Anya Kleymenova, David E. Rappoport, Francesc Rodriguez-Tours, Anatoli Segura, Alexandros Vardoulakis, Alexander Ufier, Saleem Bahaj, Kristle Cortes, Tim Eisert, Alexi Savov and seminar and conference participants at the Federal Reserve Board of Governors (Financial Stability), "Rethinking Optimal Deposit Insurance" conference (Yale), University of Kentucky Finance Conference, International Monetary Fund, BIS-CEPR-SCG-SFI Conference on Financial Intermediation (Gerzensee), Conference on "Bank liquidity" (BIS), 17th NY Fed / NYU Stern Conference on Financial Intermediation, 2024 OCC Bank Research Symposium, Banco de Portugal and CEPR Conference on Financial Intermediation (Lisbon), Annual ECB Banking Supervision Research Conference, MoFiR workshop on banking (London), CEMFI, Bundesbank Conference on Markets and Intermediaries, Fed Chicago, Barcelona Summer Forum 2025, NBER Summer Institute Corporate Finance 2025, and FIRS 2025. The views expressed in this manuscript are those of the authors and do not necessarily represent the views of the Danmarks Nationalbank, the Banca d'Italia, or the Eurosystem.

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1 Introduction

Expanding deposit insurance coverage amid crises has become a standard policy response. The United States first introduced deposit insurance during the Great Depression to stop runs on banks. During the Japanese banking crisis of the 1990s and the U.S. Savings and Loan crisis, authorities broadened coverage to stabilize their financial systems. Similarly, after the start of the 2008 Global Financial Crisis, both the European Union and the United States significantly expanded their deposit insurance schemes. More recently, the United States temporarily adopted unlimited coverage following the 2023 collapse of Silicon Valley Bank and subsequent banking sector stress. This recent episode has renewed calls for permanent unlimited deposit guarantees, driven by concerns about runs by uninsured depositors ([WSJ, 2023a,b](#)).

The primary rationale for deposit insurance rests on the historical prevalence of bank runs during financial crises — from nineteenth-century bank runs and the 1930s Great Depression ([Friedman and Schwartz, 1963](#); [Reinhart and Rogoff, 2009](#)) to the 2008 Global Financial Crisis ([Shin, 2009](#)) and the 2023 U.S. banking distress ([FDIC, 2023](#); [White, 2023](#)). Theory demonstrates that bank runs can cause even solvent but illiquid banks to fail ([Diamond and Dybvig, 1983](#)).

However, empirical evidence suggests that bank failures stem primarily from solvency problems, particularly weak fundamentals due to credit risk, rather than liquidity problems caused by runs ([Correia, Luck and Verner, 2026](#); [Baron, Laeven, Pénasse and Usenko, 2026](#)). This finding may have important implications for the costs and benefits of deposit insurance.¹ Theoretically, runs may be more likely at banks with weaker fundamentals, whether insolvent ([Allen and Gale, 1998](#)) or solvent ([Goldstein and Pauzner, 2005](#)), rather than occurring randomly due to sunspot-driven coordination failures as in [Diamond and Dybvig \(1983\)](#).

These observations motivate an important question: What do expansions of deposit insurance actually accomplish during financial crises? Despite the policy importance of this question, empirical evidence on the benefits and costs of increasing deposit insurance during banking crises is scant. Our paper examines how changes in deposit

¹[Dávila and Goldstein \(2023\)](#) formalize benefits and costs in a model of optimal deposit insurance that depends on bank fundamentals.

insurance coverage during a crisis affect the allocation of deposits across banks, particularly to **institutions** with weak fundamentals and the subsequent allocation of credit to nonfinancial corporations.

We exploit Denmark's unique institutional setting for identification, drawing on three key features: (i) comprehensive administrative datasets covering the universe of retail and corporate deposits (the deposit register) and of corporate loans (the credit register), matched with individual-, bank- and firm-level data; (ii) Denmark's shift from limited to unlimited deposit insurance following the Global Financial Crisis after Lehman Brothers' collapse in September 2008; and (iii) cross-sectional variation in bank fundamentals, exposure to uninsured depositors, and samples of households and firms with multiple banking relationships.

We measure bank weakness using the loan-to-deposit ratio as of 2007, prior to the Global Financial Crisis. Banks with **higher ratios** exhibited larger loan volume and growth during the credit boom, notably in real estate. Banks with higher loan-to-deposit ratios also relied more heavily on uninsured deposits and maintained lower capital ratios. Academic literature identifies these factors as key determinants of bank failures ([Schularick and Taylor, 2012](#); [Jordà et al., 2013, 2016](#)). This bank measure has also been used by IMF-EU-ECB rescue programs during European banking crises after 2010 for detecting risky banks and applied in **prominent research** analyzing credit supply in Denmark during the Global Financial Crisis ([Jensen and Johannesen, 2017](#)). Moreover, banks with higher loan-to-deposit ratios during the boom experienced higher loan defaults once the crisis began.

Our analysis reveals that increased deposit insurance triggers deposit reallocation from stronger to weaker banks. We first use quarterly bank-level data on deposit volumes and rates, which allow us to compare funding conditions immediately before and after the October 2008 insurance expansion. We find that, while weaker banks initially suffer funding liquidity squeezes (lower deposits and higher deposit rates) at the onset of the Global Financial Crisis, this pattern immediately reverses with unlimited deposit insurance.

To isolate deposit insurance effects from other crisis-period policies and events, we then exploit deposit balances in narrow windows around the DKK 300,000 insurance

threshold. The design compares balances just above and below the threshold – e.g. DKK 25,000 (about EUR 3,400) – within the same bank and year, so broader crisis shocks to a bank are held fixed while insurance coverage changes discretely at the threshold. Under limited insurance, weak banks lose deposits just above the threshold relative to deposits just below it. Once coverage becomes unlimited, this differential outflow progressively disappears as weak banks start gaining large (previously uninsured) deposits (i.e., above the threshold), showing that the insurance reform drives the reallocation of large deposits **for** weak banks.

As individuals with deposits just above versus just below the insurance limit may be different, and hence they may react differently to the shock, we exploit the sample of individuals with at least two banks and employ individual fixed effects. We find that the same person simultaneously withdraws more deposits from weaker banks, particularly above the insurance limit, under limited deposit insurance. Conversely, under unlimited insurance, the same individual moves large deposits (above the previous insurance threshold) from stronger to weaker banks. Moreover, corporate depositors also exhibit similar patterns, though firms do not return large deposits to weak banks under unlimited insurance as retail depositors do. Furthermore, we find stronger effects among high net-worth depositors, including both wealthy individuals and firms with high equity.

One of the concerns might be that depositor response is a result of weak banks wanting to attract fewer deposits when deposit insurance is limited. Using deposit pricing, along with deposit volume, we find evidence consistent with the results being driven by depositors' response rather than banks. In particular, weaker banks raise deposit rates to attract funding in 2008 under limited deposit insurance, reflecting their greater deposit funding difficulties. However, these higher deposit rates gradually decrease over time under unlimited insurance. The combination of unlimited deposit insurance and higher deposit rates make weak banks more attractive, enabling them to halt deposit losses and gain large deposits above the previous insurance limit.

To understand better the implications of these deposit dynamics, we examine how deposit insurance affects bank lending behavior. Banks with weak fundamentals may change credit supply in crisis times for reasons unrelated to deposit insurance. How-

ever, as discussed earlier, we find that banks with weaker fundamentals and a higher share of uninsured deposits are more affected by deposit insurance reform. Therefore, to trace how deposit shifts affect lending, we focus on weak banks with a high concentration of uninsured depositors, which makes them more exposed to deposit insurance changes. That is, our identification strategy for the credit analysis combines two measures, our pre-crisis weakness indicator (loan-to-deposit ratios) with a bank-level measure of uninsured depositor concentration prior to the crisis. Moreover, following Khwaja and Mian (2008), we employ firm fixed effects and analyze companies in the credit register that maintain relationships with at least two banks, enabling us to control for firm-specific fundamentals, including firm-level credit demand.

We find that banks exposed to the insurance reform cut credit supply during the crisis under limited deposit insurance but expand credit supply under unlimited insurance. However, this credit supply expansion by exposed banks with unlimited insurance disproportionately benefits risky firms, measured either by ex-ante risk indicators or by ex-post outcomes (defaults).² To gauge the magnitude, among banks most exposed to the reform, the share of new credit going to ex-ante riskier firms doubles once all deposits are insured. Moreover, despite this increased risk-taking, exposed banks do not charge higher loan rates to riskier borrowers.

Finally, we examine banks that ultimately failed during 2009–2015 as an ex-post benchmark of fundamentally weak banks as well as to understand the implications of our previous results on deposits and lending. We analyze whether pre-crisis bank characteristics and crisis-period deposit flows predict subsequent bank failure.

We find that, under limited insurance, deposit outflows during the crisis predict future bank failure, both in general and when driven by our pre-crisis weakness measure. After unlimited guarantees were introduced, deposit inflows instead predict failure, and more strongly for flows driven by pre-crisis weakness. The relationship between deposit flows and future bank failure thus reverses completely, so that banks

²We also find that, once deposit insurance is unlimited, weaker banks (higher loans to deposits) increase credit supply to ex-ante riskier borrowers and to borrowers with higher ex-post defaults. Moreover, as we wrote above, these effects are enhanced for those weak banks with higher shares of uninsured deposits, i.e. weak banks more exposed to the deposit insurance reform. In addition, despite that we find that these weaker banks with higher uninsured deposits increase credit supply in 2009 but decrease it in 2008, we find that, on average, weaker banks decrease it in both years.

attracting the most deposits become the most likely to fail ex-post. Banks in the *top* quartile of predicted 2008:Q4 deposit growth failed **at 48%** over 2009–2015, against just 15% in the bottom three quartiles, more than three times as likely. Our pre-crisis weakness measure predicts bank failures substantially better than realized deposit flows or standard bank risk measures, such as capital ratios and size. Together with the deposit and lending results above, these findings suggest that expanded deposit insurance redirected funds toward fundamentally weak banks, which then increased credit supply to very risky firms.

In sum, our findings demonstrate that deposit insurance prevents weaker banks from further losing deposits during financial crises, enabling continued credit supply to firms. However, increasing deposit insurance also facilitates deposit reallocation from stronger to weaker banks, with weaker institutions subsequently increasing credit supply to ex-ante high-risk firms that experience higher ex-post failure rates. Consistently, banks that gained the most deposits under unlimited deposit insurance are themselves more likely to fail ex-post, particularly those banks that were weak ex-ante.

Contribution to the literature. Although deposit insurance coverage is typically expanded during crises when bank solvency and gambling-for-resurrection concerns are most acute, existing literature primarily examines moral hazard issues associated with deposit insurance in normal times ([Demirgüç-Kunt and Detragiache, 2002](#); [Demirgüç-Kunt and Huizinga, 2004](#); [Beck and Laeven, 2008](#); [Calomiris and Jaremski, 2019](#)). We contribute to this literature by analyzing the effects of deposit insurance expansion specifically during crisis periods. In addition, our analysis goes beyond previous work by examining both deposit reallocation and its associated credit supply and risk-taking effects, while strengthening the causal identification.

Previous studies using granular data demonstrate that deposit insurance reduces uninsured deposit runs, but these analyses are single-bank case studies ([Iyer and Puri, 2012](#); [Iyer et al., 2016](#); [Martin et al., 2026](#)). Using complete deposit register data, we extend this literature by showing that deposit insurance generates deposit reallocation from stronger to weaker banks, enabling weak banks with high uninsured depositor concentrations to expand their credit supply. Additionally, we provide novel evidence

on corporate depositor behavior, an area less studied empirically in the literature due to data limitations (Chernykh and Mityakov, 2022).

Our findings reveal that increased deposit insurance coverage benefits more banks with weaker solvency by alleviating liquidity constraints, consistent with Correia, Luck and Verner (2026) and Baron, Laeven, Pénasse and Usenko (2026). Importantly, we document misallocation through deposit flows from strong to weak banks and increased credit supply to very risky firms by the banks that benefit most from deposit insurance expansion (Hsieh and Klenow, 2009; Acharya et al., 2022). Strikingly, many banks that receive deposits due to deposit insurance expansion eventually fail, suggesting that deposit insurance may contribute to the persistence of very weak banks.

Our empirical results also contribute to the theoretical debate on deposit insurance’s role in different types of bank runs (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005) and its impact on market discipline (Calomiris and Kahn, 1991; Diamond and Rajan, 2001; Rochet and Vives, 2004; Dávila and Goldstein, 2023). We find evidence consistent with bank runs driven by weak fundamentals (Gorton, 1988; Goldstein and Pauzner, 2005; Allen and Gale, 1998, 2004) due to credit risk concerns (Temin, 1976; Wicker, 1996; Calomiris and Mason, 2003; Admati and Hellwig, 2014; Gennaioli and Shleifer, 2018). Correspondingly, we show that depositor discipline is weakened by higher deposit insurance limits.

2 Institutional Setting and Data

This section describes the institutional setting, data, and key variables used in the empirical analysis. Denmark combines a crisis-time shift from limited to unlimited deposit insurance with granular administrative data on deposits, bank balance sheets, firms, and credit relationships. These features allow us to compare deposit and credit allocation across banks that differ in pre-crisis balance sheet strength and in their exposure to uninsured deposit funding.

2.1 Denmark's Deposit Insurance Reform

Denmark's policy response to the Global Financial Crisis provides the institutional variation for our research question. Leading up to 2008, Danish banks funded a domestic real estate boom largely through wholesale foreign borrowing rather than direct exposure to U.S. subprime assets. When international funding markets seized following Lehman Brothers' collapse in September 2008, Danish banks experienced funding pressure despite limited holdings of toxic mortgage-backed securities (IMF, 2011; Jensen and Johannesen, 2017). Falling domestic real estate prices further weakened bank balance sheets.

Facing funding stress, the Danish government introduced crisis measures in October 2008 under "Bank Rescue Package I." The package guaranteed all bank deposits in Denmark in full, suspending the existing insurance limit of DKK 300,000 (EUR 40,000). The government announced a scheduled expiry of 30 September 2010, so the unlimited deposit guarantee's end date was public knowledge from the outset.³

The timing creates two crisis periods with different deposit insurance regimes. In 2008:Q3, banks faced funding stress while deposits were insured only up to DKK 300,000. From 2008:Q4 through September 2010, the crisis continued under unlimited coverage. We therefore observe an expansion of deposit insurance during an active financial crisis.

2.2 Administrative Data and Sample Construction

The Danish administrative registers allow us to measure the reallocation of deposits and credit within and across banks in Denmark. This matters because the paper studies both the immediate deposit response to higher insurance coverage and the subsequent lending behavior of banks. Our analysis combines matched administrative datasets from Statistics Denmark. The account-level information on deposits and credit covers the period from 2004 through 2010 and is linked to bank failure and firm default outcomes observed through 2015. The bank sample includes 94 banks super-

³The agreement was reached on 5 October 2008 and enacted into law on 10 October 2008. The legislation allowed for prolongation if financial stability so required, but no prolongation occurred: the guarantee lapsed as scheduled. Abildgren and Thomsen (2011) and Rangvid et al. (2013) provide further details on Denmark's crisis response.

vised by the Danish Financial Supervisory Authority as of 2007.

The deposit register covers all deposit account types held by individuals and firms at Danish financial institutions through mandatory year-end reporting to the tax authority (SKAT). This reporting structure limits selection concerns and lets us study not only withdrawals from individual banks but also the reallocation of deposits across the banking system. We observe balances and annual interest payments for all deposit account types, including checking accounts and term deposits, but we cannot distinguish between account types. As in the United States, Danish deposit insurance applies at the depositor-bank level, so depositors can increase insured balances by spreading deposits across banks but not by splitting deposits across accounts within the same bank. We therefore aggregate account-level deposits and interest payments to the depositor-bank-year level. At this level, we mark a depositor's balance as uninsured in a given year when it exceeds DKK 300,000 at that bank.

The credit register has the same structure and covers all non-mortgage lending agreements between banks and nonfinancial firms, including term loans and credit lines.⁴ We observe outstanding credit balances and interest payments, but we cannot distinguish between credit products. We aggregate lending agreements to the bank-firm-year level, which allows us to compare credit supplied by different banks to the same firm.

Since contractual interest rates are inconsistently reported for tax purposes, we compute effective rates from observed interest payments and outstanding balances. For deposits, we calculate the effective annual rate as total deposit interest payments in year t divided by the average deposit balance in year t . We approximate average balances using the mean of balances at the start and end of year t , measured as end-of-year balances in $t - 1$ and t . We construct loan rates analogously, using loan interest payments divided by average outstanding credit. Both deposit and loan rates are expressed as decimals throughout the paper (e.g., 0.01 corresponds to a 1 percentage point rate).

To focus on economically meaningful relationships, we impose several sample re-

⁴In Denmark, mortgages are issued by specialized mortgage institutions under distinct regulations, which we exclude from our analysis.

restrictions on firms. We include only nonfinancial firms classified as economically active by Statistics Denmark, excluding cooperatives, NGOs, sole proprietorships, and other non-standard corporate forms. We further require annual sales of at least DKK 75,000 (EUR 10,000) and at least one employee to separate operating firms from personal financial arrangements. These restrictions yield a final sample of 64,859 nonfinancial firms.

We link these data to individual demographics, wealth, income, and geographic location from tax records; bank balance sheets, income statements, and regulatory metrics from supervisory data; and corporate legal status, founding year, employment, and financial statements from business registers.

2.3 Bank Weakness and Reform Exposure Measures

To document how changes in the deposit insurance limit reallocate deposits and credit across banks, we exploit variation in two pre-crisis bank characteristics. The 2007 loan-to-deposit ratio is our main measure of banks with weak fundamentals. The share of uninsured deposits in 2007 measures how strongly a bank's deposit base was affected by the removal of the deposit insurance limit.

The loan-to-deposit ratio is relevant for several reasons. It captures credit growth that outpaced deposits, large loan books relative to stable deposit funding, and reliance on less stable non-deposit funding. These features left banks with higher loan-to-deposit ratios more vulnerable when wholesale funding markets froze. The ratio also tracks credit risk in our data: banks with higher loan-to-deposit ratios lent more during the boom, especially to firms in real estate and construction, sectors hit hard by the downturn. Section 3 documents these patterns and shows that these banks had weaker loan portfolios, higher loan losses, and a higher probability of failure between 2009 and 2015, which is why we interpret a higher loan-to-deposit ratio as a proxy for weaker bank fundamentals.

The loan-to-deposit ratio is also a policy-relevant financial stability measure. IMF-EU-ECB rescue programs during European banking crises after 2010 used banks' loan-to-deposit ratios to identify institutions with excessive reliance on less stable non-deposit funding (IMF, 2011). In related Danish evidence, Jensen and Johannesen (2017)

classify banks with high 2007 loan-to-deposit ratios as crisis-exposed and show they cut household lending relative to other banks.

The loan-to-deposit ratio measures general bank weakness, but banks also differed in how directly the reform affected their funding. We capture this second dimension with the 2007 uninsured deposit share, defined as the fraction of retail and nonfinancial corporate deposits above the DKK 300,000 insurance limit. When the limit was removed, banks with a high uninsured deposit share saw much of their deposit base shift from uninsured to fully guaranteed. This bank-level exposure measure matters most on the lending side. On the deposit side, comparing balances just above and just below the DKK 300,000 threshold identifies the reform's effect on deposit flows. On the lending side, no comparable threshold identifies the reform's effect on credit supply, so we instead measure each bank's exposure to the reform with its uninsured deposit share, especially in combination with a higher loan-to-deposit ratio.

2.4 Pre-Crisis Sample Characteristics

Table [A.1](#) presents descriptive statistics as of December 2007, before the crisis unfolded. The table shows the pre-crisis variation used in the empirical analysis: banks differ in balance-sheet strength and exposure to uninsured deposits, while many depositors and firms maintain relationships with multiple banks.

Panel A describes the bank sample. Banks differ substantially in size, with assets ranging from DKK 620 million (EUR 83 million) at the 10th percentile to DKK 33.8 billion (EUR 4.5 billion) at the 90th percentile. The loan-to-deposit ratio averages 1.17 with a standard deviation of 0.37, consistent with large differences in pre-crisis bank fundamentals and funding structures, including reliance on non-deposit funding. The average bank has 42% of deposit volume above the DKK 300,000 insurance threshold. Removing the insurance limit therefore affected a large part of bank funding. Finally, 22% of banks fail between 2009 and 2015, giving us an ex-post measure of bank weakness.

Panels B and C show that uninsured balances are concentrated among larger depositors but appear in both retail and corporate deposit markets. Six percent of individuals hold deposits above the insurance limit, compared with 62% of firms. These unin-

sured balances are important at the bank level, as shown by the 42% uninsured deposit share in Panel A. Multi-bank relationships are also common enough for within-depositor comparisons: 18% of individuals and 64% of firms hold deposits at more than one bank.

Panel C also describes the credit relationships used to study lending. Forty-eight percent of firms borrow from at least two banks, which supports firm fixed-effect tests of credit supply. To study whether the reform changes how banks allocate credit across firms of different riskiness, we use both ex-ante and ex-post measures of borrower risk. The ex-ante measure is a firm risk index that combines three accounting drivers of default risk: leverage, profitability, and earnings volatility. It equals $(\text{Leverage} - \text{RoA})$ multiplied by the volatility of RoA over the current and previous two years, so it rises with leverage and earnings volatility and falls with profitability.⁵ The index averages 0.05, with a sizeable standard deviation of 0.17. The ex-post measure is firm default: among firms observed in 2007, 26% default at some point between 2009 and 2015. In line with these risk measures, firm capitalization also varies substantially: the (book) equity ratio averages 0.37, with a 10th-to-90th percentile range from 0.09 to 0.71.

3 Bank Weakness and Crisis Funding Stress

This section validates our use of pre-crisis loan-to-deposit ratios as a measure of bank weakness and examines how banks with different fundamentals responded to the change in deposit insurance during the Global Financial Crisis. We show that banks with higher loan-to-deposit ratios entered the crisis with weaker fundamentals and experienced greater distress when funding markets seized. We then show that their funding conditions changed sharply around the deposit insurance reform.

⁵We construct an accounting-based index rather than use Altman's z-score (Altman, 1968) or a market-based distance to default (Bharath and Shumway, 2008), which are infeasible here because most Danish firms are privately held and lack the market values and accounting inputs these measures require; Altman and Sabato (2007) likewise build accounting-based default scores for private firms. Appendix Table A.7 shows the index is significantly higher for firms that subsequently default, supporting its use as an ex-ante risk ranking.

3.1 Loan-to-Deposit Ratio and Bank Weakness

We establish that banks with higher loan-to-deposit ratios in 2007 were weaker institutions through three complementary analyses: cross-sectional bank characteristics, pre-crisis loan-portfolio composition, and crisis-period loan losses. Panels A and B of Figure 1 study the correlation between banks' loan-to-deposit ratios and various bank characteristics. Unless explicitly stated otherwise, all variables in these panels are measured in 2007. To do so, we regress each characteristic on the loan-to-deposit ratio:

$$y_b = \alpha + \beta \text{Loan/Deposits}_b + \gamma X_b + \epsilon_b, \quad (1)$$

where y_b denotes a bank characteristic or log loan-portfolio amount. The baseline specification includes only the loan-to-deposit ratio; the controlled specification adds log assets, the Tier 1 capital ratio, and an indicator for the six largest banks, excluding a control when it is the dependent variable.

Panel A shows that banks with higher loan-to-deposit ratios were not simply banks with more fragile funding structures. They had faster credit growth during 2004–2007, higher real-estate credit shares, greater uninsured-deposit exposure, and lower capital ratios. They also subsequently experienced larger loan losses and were more likely to fail between 2009 and 2015. Taken together, these correlations link the ratio to weak pre-crisis fundamentals and funding structures: rapid credit expansion, exposure to real estate and uninsured deposits, thinner capital, and eventual distress.

Panel B provides additional evidence on the loan portfolios of banks with higher loan-to-deposit ratios using matched credit-register and firm-level data. These banks entered the crisis with larger loan portfolios, especially in nonfinancial corporate lending, credit to firms in the real-estate and construction industries, and credit to firms with high values of the ex-ante firm risk index. These associations remain positive after adding additional bank-level controls. The figure therefore shows that the loan-to-deposit ratio captures both the funding side of pre-crisis expansion and the asset-side risk associated with that expansion.

Next, we examine whether these portfolio differences translated into higher loan losses once the crisis hit. Panel C of Figure 1 presents coefficient estimates with 95%

confidence intervals from bank-year regressions of loan losses scaled by total assets on banks' 2007 loan-to-deposit ratios interacted with year indicators. The specification includes bank and year fixed effects and interacts year indicators with de-meaned 2007 controls for the Tier 1 capital ratio and log assets, plus an indicator for the six largest banks. Coefficients are normalized relative to 2006, and standard errors are clustered at the bank level.

The figure shows that before 2008, banks with higher loan-to-deposit ratios did not experience significantly different loan losses, with point estimates close to zero. However, once the crisis began, banks with higher loan-to-deposit ratios experienced significantly larger loan losses from 2008 to 2015. This pattern supports the interpretation that pre-crisis lending expansion and portfolio composition captured weak fundamentals due to credit risk, with effects persisting well beyond the acute phase of the GFC. The timing and persistence of these losses validate our use of the loan-to-deposit ratio as an indicator of bank vulnerability entering the crisis.

3.2 Quarterly Funding Stress Around the Reform

Having established that banks with higher loan-to-deposit ratios entered the crisis with weaker fundamentals and experienced greater loan losses in the subsequent years, we next ask whether bank-level funding conditions changed around the October 2008 deposit insurance expansion in a way consistent with weaker banks receiving funding relief. This exercise is useful because quarterly supervisory data let us observe funding conditions immediately before and after the reform, whereas our granular deposit-account data are annual. We estimate cross-sectional regressions that compare banks with different 2007 loan-to-deposit ratios in 2008:Q3, before the deposit insurance expansion, and in 2008:Q4, after unlimited coverage was introduced. Because these regressions are cross-sectional and coincide with a turbulent crisis period, we interpret them as suggestive higher-frequency evidence rather than causal estimates.

Table 1 examines quarter-on-quarter changes in log deposits (Panel A) and deposit rates (Panel B) for banks with higher loan-to-deposit ratios in 2007. The specifications include bank controls and region fixed effects. During 2004:Q1–2007:Q4 and in the first two quarters of 2008, deposit flows and deposit-rate changes do not differ significantly

across banks with different loan-to-deposit ratios. In 2008:Q3, at the onset of the crisis while deposit insurance was still capped at DKK 300,000, banks with higher loan-to-deposit ratios experienced deposit outflows relative to lower-ratio banks and raised deposit rates.

After unlimited coverage was introduced, these funding-pressure indicators reverse. In 2008:Q4, a one-standard-deviation increase in the loan-to-deposit ratio is associated with roughly 6 log points higher deposit growth (0.16×0.37). Relative to the 2008:Q3 coefficient, the Q4 deposit-growth coefficient rises by about 9 log points for a one-standard-deviation higher loan-to-deposit ratio (0.23×0.37), implying a shift from relative outflows to relative inflows. Deposit-rate changes also reverse: after banks with higher loan-to-deposit ratios raised rates in 2008:Q3, the Q4-minus-Q3 difference implies that these rate increases abate by about 41 basis points (-1.10×0.37) for a one-standard-deviation higher loan-to-deposit ratio once unlimited coverage is introduced.⁶ The changes in the estimated coefficients between 2008:Q3 and 2008:Q4 for both deposit volumes and rates are statistically significant at the 1% level.

The reversal does not depend on dating the loan-to-deposit ratio at the end of 2007. Columns 6–8 re-estimate the 2008:Q3 and 2008:Q4 regressions with the ratio measured in 2008:Q2 or in 2008:Q3, so that it reflects each bank’s funding position immediately before the outcome quarter. Banks with higher ratios again show deposit outflows and rate increases relative to lower-ratio banks in 2008:Q3, and both patterns reverse in 2008:Q4. Every coefficient is significant at the 10% level or better, and the Q4-minus-Q3 difference at the 2008:Q2 dating is significant at the 1% level in both panels.

Overall, the quarterly evidence suggests that banks with higher loan-to-deposit ratios faced funding pressure under limited insurance, but that this pressure eased once large deposit balances were fully guaranteed. We next use micro-level deposit data to test the same mechanism with designs that isolate changes around the insurance threshold and within depositors who hold accounts at multiple banks.

⁶Observation counts differ across columns because some banks do not report all quarterly deposit or rate variables. Appendix Table A.2 restricts the sample to banks without missing observations throughout 2007:Q1–2008:Q4, and confirms the Q3–Q4 reversal.

4 Deposit Insurance and Deposit Reallocation

This section examines how deposit insurance shaped the allocation of deposits across banks during the Global Financial Crisis (GFC). The quarterly evidence already showed weaker banks losing deposits while coverage was limited and regaining them once it became unlimited. This reversal is consistent with market discipline: uninsured depositors had reason to withdraw from weaker banks only while balances above DKK 300,000 were at risk. The bank-level data cannot pin down this mechanism, however, because they neither isolate the uninsured balances on which discipline operates nor separate the reform from other crisis shocks. We therefore use micro-level deposit data to ask directly whether large balances left weaker banks while coverage was limited, and whether this reallocation stopped once coverage became unlimited. We study both household deposits and deposits held by nonfinancial companies.

Two identification challenges guide the analysis. First, weak banks were exposed to many crisis-period shocks and policy interventions that could affect deposit growth independently of deposit insurance. We address this concern by comparing deposit balances just above and just below the DKK 300,000 insurance limit: coverage changes discontinuously at the limit, but other crisis shocks have no reason to affect a bank's deposits differently just above versus just below it. Second, depositors may have selected across banks before the crisis. More attentive or sophisticated depositors might have held deposits at weaker banks and withdrawn during the crisis for reasons unrelated to the reform. We address this concern using depositor-bank-level regressions for depositors with multiple bank relationships, comparing the same depositor's balances across banks.

4.1 Deposit Flows around the Insurance Limit

We begin with the distribution of retail deposit balances around the DKK 300,000 insurance threshold. Figure 2 plots binned empirical distributions of deposit balances at two aggregation levels, with each point reporting the number of observations in a DKK 1,000-wide bin. Panel A shows the distribution at the depositor-bank level, the level at which deposit insurance applies. Balances bunch just below the limit, indicating that depositors manage balances within each bank to remain insured. This bunch-

ing tracks the insurance regime over time: it is minimal in 2007, rises sharply in 2008 as the crisis takes hold while coverage remains capped, and subsides in 2009–2010 once coverage becomes unlimited. Panel B instead sums each individual’s deposits across all banks, so each range now counts individuals. Had depositors simply withdrawn funds from the banking system, this individual-level distribution would bunch as well; instead it does not, consistent with depositors spreading balances across banks to stay insured within each bank.

The bunching evidence motivates a regression analysis of whether deposit balances changed differentially in narrow windows around the DKK 300,000 threshold during the crisis. In this threshold comparison, we analyze both retail and nonfinancial corporate deposits. Following [Iyer et al. \(2019\)](#), we aggregate deposits into bins of DKK 25,000 at the bank-year level. This construction is useful because every depositor-bank balance above the insurance limit contains both insured and uninsured amounts; decomposing balances into range bins lets us compare deposit volumes immediately below and above the insured portion of the balance. To construct the data, we split each depositor-bank balance into range bins and aggregate these amounts across all depositors, banks, and years. For example, a depositor-bank balance of DKK 210,000 contributes DKK 25,000 to each of the first eight bins (0–25K, 25–50K, ..., 175–200K) and DKK 10,000 to the ninth bin (200–225K). This approach produces a bank-bin-year panel, which we use to estimate the following regression model:

$$\log(\text{Deposits})_{btk} = \alpha_{kt} + \alpha_{bt} + \sum_t \beta_t (\eta_t \times \text{Loan/Deposits}_{b,2007} \times \text{Above } 300K_{kt}) + \epsilon_{btk} , \quad (2)$$

where the dependent variable $\log(\text{Deposits})_{btk}$ denotes the logarithm of total retail and corporate deposits held by bank b in year t within bin k . The indicator $\text{Above } 300K_{kt}$ equals one for bins above the DKK 300,000 deposit insurance threshold in year t , and η_t denotes a full set of year indicators. Lower-order terms are included where separately identified. The coefficients of interest, β_t , capture the differential change in deposits above versus below the threshold at banks with higher 2007 loan-to-deposit ratios, normalized relative to 2006. A negative coefficient implies that banks with higher loan-to-deposit ratios lost deposits just above the insurance limit relative to deposits

just below it, compared with banks with lower loan-to-deposit ratios.

The specification includes bin-year fixed effects (α_{kt}), which absorb time-varying heterogeneity in deposit volumes across bins, and bank-year fixed effects (α_{bt}), which absorb all time-varying deposit dynamics within each bank-year. This saturated fixed-effects structure ensures that identification comes solely from within-bank variation across threshold bins.

Figure 3, Panel A shows the dynamic coefficients from this specification, estimated for progressively narrower windows around the insurance threshold. The narrowest window uses one DKK 25,000 bin on each side of the threshold, approximately EUR 3,400 above and below the insurance limit. Prior to 2008, deposit levels above and below the threshold evolved in parallel, with year-specific coefficients (measured relative to the omitted 2006 baseline) close to zero and statistically insignificant, supporting the validity of our identification strategy. In 2008, however, the coefficient becomes large and negative; because year-end 2008 balances include roughly two months of unlimited-coverage exposure, this estimate is best read as a lower bound. A one-standard-deviation increase in the loan-to-deposit ratio is associated with about a 3 log point decline in deposits above versus below the threshold in the narrowest DKK 25,000 window (-0.08×0.37). This substantial difference indicates that depositors reduced uninsured balances at banks with higher loan-to-deposit ratios during the crisis period when deposit insurance remained limited.

After the reform, this differential outflow of above-threshold deposits at banks with higher loan-to-deposit ratios largely disappears. In 2009, when unlimited deposit insurance was in effect for the full year, the negative interaction is close to zero, and further attenuated when combined with the lower-order interaction effects reported in Table 2. The same pattern appears in 2010. Thus, the threshold-level evidence points to withdrawals of uninsured balances from banks with higher loan-to-deposit ratios under limited coverage, followed by a sharp weakening of this differential outflow once coverage became unlimited.

We use a placebo threshold to test whether the result reflects the insurance limit rather than a generic response around round-number deposit balances. Figure 3, Panel A repeats the same dynamic framework in narrow windows around DKK 250,000,

an arbitrary cutoff below the regulatory threshold. Around this placebo threshold, we do not find differential withdrawals from banks with higher loan-to-deposit ratios just above versus just below the cutoff. The contrast with the DKK 300,000 insurance limit supports the interpretation that deposit reallocation is tied to the regulatory threshold.

Figure 3, Panel B repeats the threshold-bin analysis using the annual change in log deposits as the dependent variable. The estimates through 2008 mirror Panel A: pre-crisis coefficients are close to zero, the 2008 interaction is negative around the insurance threshold, and the placebo threshold shows no comparable movement. In 2009, however, the interaction becomes positive and statistically significant, in contrast to the near-zero 2009 effect in Panel A. Because the main log-deposit specification already identifies relative deposit growth through bank-year and bin-year fixed effects, this positive estimate means that deposit growth above the former insurance limit increased at banks with higher loan-to-deposit ratios after coverage became unlimited, rather than merely returning to the growth rate of deposits below the limit.

4.2 Deposit Reallocation Within Depositors

We next turn from evidence in narrow windows around the insurance threshold to deposit growth at the depositor-bank level. Cross-bank regressions, reported in Appendix Table A.3, show the same timing as the funding-pressure evidence above: in 2008, deposit growth falls at weak banks with large previously uninsured deposit shares, and this differential withdrawal pattern disappears in 2009 after unlimited coverage is introduced. These patterns are useful descriptive evidence, but they may reflect depositor sorting across banks. Banks with higher loan-to-deposit ratios may attract depositors with different risk preferences, liquidity needs, or socioeconomic characteristics, potentially confounding the relationship between bank fundamentals and depositor behavior during crises (Kashyap et al., 2002; Gatev and Strahan, 2006; Ippolito et al., 2016). We therefore restrict the main depositor-level analysis to depositors with multiple bank relationships and exploit within-depositor variation in withdrawal behavior, comparing how the same depositor adjusts balances across banks with different pre-crisis loan-to-deposit ratios. We estimate the following depositor-

bank-year regression in each of the years 2008 and 2009:

$$\begin{aligned} \Delta \log(\text{Deposits})_{ibt} = & \alpha_i + \alpha_b + \beta_1 \text{Loan/Deposits}_{b,2007} \times \text{Above 300K}_{i,b,t-1} \\ & + \beta_2 \text{Above 300K}_{i,b,t-1} + \epsilon_{ibt} \end{aligned} \quad (3)$$

where $\Delta \log(\text{Deposits})_{ibt}$ denotes the annual change in log deposits for depositor i at bank b . The variable $\text{Loan/Deposits}_{b,2007}$ is bank b 's loan-to-deposit ratio at the end of 2007, and $\text{Above 300K}_{i,b,t-1}$ is an indicator equal to one if depositor i 's deposits at bank b exceeded DKK 300,000 in the previous year. The interaction term tests whether depositors with balances above the insurance threshold withdrew more from banks with higher pre-crisis loan-to-deposit ratios. Depositor fixed effects α_i absorb all time-invariant depositor characteristics, such as risk preferences, sophistication, and baseline liquidity needs. Bank fixed effects α_b absorb time-invariant bank characteristics that affect deposit growth across all relationships at the same bank. Standard errors are double-clustered at the depositor and bank level.

Panel A of Table 3 presents the regression results for individual depositors, corporate depositors, and the pooled sample of retail plus corporate depositors. For individual depositors, the coefficient on the interaction between the loan-to-deposit ratio and the above-threshold indicator is negative and significant in 2008. A one-standard-deviation increase in the loan-to-deposit ratio is associated with about 4.4 percentage points lower deposit growth in depositor-bank balances that exceeded the insurance threshold. This response is consistent with uninsured retail depositors monitoring banks' fundamentals and systematically withdrawing large balances especially from weak banks.⁷

This pattern reverses in 2009, when deposit insurance covered all balances. The interaction term for individual depositors turns positive and statistically significant. The coefficient of 0.05 implies that a one-standard-deviation increase in the loan-to-deposit ratio is associated with roughly 1.9 percentage points higher deposit growth

⁷Public awareness of bank fundamentals at the time was aided by widespread media coverage that translated technical risk metrics into intuitive rankings and simplified indicators. Notable examples include Berlingske Business's "Banking Risk Index" that explicitly ranked institutions by risk factors (April-September 2008), Politiken's coverage of Moody's downgrades ("Moody's rammer danske banks," September 17, 2008), and extensive television coverage on TV2 and DR1 that explained banking fundamentals to general audiences.

in depositor-bank balances that previously exceeded the insurance limit. For retail depositors, unlimited coverage therefore turns large-deposit flows back toward weaker banks.

We extend our analysis to corporate deposits by nonfinancial companies using the same identification strategy. In contrast to retail depositors, corporate depositors are typically considered more sophisticated and run-prone, making them a natural comparison group. Panel A of Table 3 also presents results for firm-bank-level deposit changes. The patterns are similar to those observed for individual depositors under limited coverage. In 2008, corporate depositors with uninsured balances withdrew more from banks with higher loan-to-deposit ratios; the coefficient is larger in magnitude than for individuals, although estimated less precisely. Unlike retail depositors, corporate depositors do not clearly return large deposits to banks with higher loan-to-deposit ratios in 2009: the corresponding coefficient is close to zero.

Columns 5–6 of Panel A of Table 3 pool individuals and firms to confirm that the average depositor with multiple bank relationships withdraws large balances from banks with higher loan-to-deposit ratios in 2008 and returns them in 2009, with the interaction significant in both years. This reversal is consistent with expanded coverage weakening market discipline: once fully insured, depositors no longer monitor bank risk and move large balances back toward weak banks. One channel for this return runs through deposit rates: banks with higher loan-to-deposit ratios may attract these balances by offering more competitive rates that fully insured depositors no longer weigh against risk, a channel we examine in subsection 4.3.

Heterogeneity by Exposure and Depositor Characteristics. To provide further evidence on the mechanism behind targeted deposit withdrawals at weaker banks, we examine heterogeneity across bank exposure and depositor characteristics. We augment the specification in equation (3) with a triple interaction term between the loan-to-deposit ratio, the above-threshold indicator, and one moderator at a time. Panel B of Table 3 reports these estimates, starting with banks' uninsured-deposit share, then retail depositor wealth, and finally firms' equity ratio.

Columns 1–6 show that the deposit-flow effects are concentrated where all three

margins of the triple interaction coincide: they are strongest for above-threshold (previously uninsured) balances, held at banks with higher loan-to-deposit ratios that also had a larger pre-crisis share of uninsured deposits. Among retail depositors, and in the pooled sample of retail and corporate depositors, the triple interaction is negative in 2008 and positive in 2009. In the sample of corporate depositors, the estimates have the same signs but are less precise. This pattern links the within-depositor reallocation results directly to banks' exposure to the deposit insurance reform: banks with more uninsured deposits experience stronger withdrawals from uninsured depositors under limited coverage, and this sensitivity reverses once large deposit balances become fully guaranteed.

Columns 7–8 show that wealthier individual depositors react more strongly around the reform. Wealth is a natural proxy for depositor sophistication, so these depositors may monitor bank risk more closely. The triple interaction with standardized 2007 wealth is negative in 2008 and positive in 2009, consistent with high-net-worth retail depositors driving the withdrawal of large balances from weaker banks under limited insurance, and the return of large retail deposits under unlimited insurance.

Columns 9–10 show the corresponding heterogeneity for firms. The triple interaction with firms' 2007 equity ratio is negative and significant in 2008, but close to zero in 2009. This pattern is consistent with better-capitalized firms being more able to reallocate liquidity away from weak banks when their uninsured balances are exposed, without a comparable return of large corporate deposits once coverage becomes unlimited.

4.3 Deposit Pricing Around the Reform

Deposit pricing helps distinguish depositor-side withdrawal pressure from banks' demand for deposit funding. If outflows from weak banks reflected these banks reducing their demand for deposits, we would not expect them to raise deposit rates when large balances were leaving. If instead depositors withdrew because uninsured balances exposed them to bank risk, weak banks should pay higher rates to retain funding.

We therefore examine deposit rates around the reform. For each bank-year, we compute the average deposit rate across the bank's retail and nonfinancial corporate

deposit relationships. Panel A of Figure 4 compares this average rate for banks with above-median versus below-median loan-to-deposit ratios in 2007. The two groups offered similar rates before 2007, but a sharp divergence emerges during the crisis. Banks with higher loan-to-deposit ratios raised rates substantially in 2008, precisely when large depositors were withdrawing from these institutions. The gap narrows after unlimited insurance is introduced, but banks with higher loan-to-deposit ratios continue to offer higher deposit rates through 2009, before the gap closes by 2010.

While informative, this descriptive evidence does not adjust for other pre-crisis bank characteristics and aggregate shocks during the crisis. We therefore quantify the relationship using a panel regression that traces the time-varying relationship between the 2007 loan-to-deposit ratio and subsequent deposit pricing:

$$\text{Rate}_{bt} = \alpha_b + \alpha_t + \sum_t \beta_t (\eta_t \times \text{Loan/Deposits}_{b,2007}) + \sum_t \gamma_t (\eta_t \times X_b) + \epsilon_{bt} \quad (4)$$

where Rate_{bt} is the average deposit rate at bank b in year t , computed across retail and nonfinancial corporate depositors. The coefficients of interest, β_t , measure how deposit rates at banks with higher 2007 loan-to-deposit ratios differ in each year, normalized relative to 2006. The vector X_b contains 2007 bank characteristics interacted with year indicators: de-meaned Tier 1 capital, de-meaned log assets, and an indicator for the six largest banks. Bank fixed effects α_b absorb time-invariant differences in deposit pricing across banks, while year fixed effects α_t absorb aggregate movements in deposit rates. Standard errors are clustered at the bank level.

Panel B of Figure 4 plots the estimated β_t coefficients. The coefficients remain near zero before 2008, but become large and statistically significant in 2008, when funding pressure on banks with higher loan-to-deposit ratios was most acute. In 2009, after unlimited insurance is introduced, the rate gap narrows but remains positive and statistically significant. By 2010, the pricing gap closes. Because account and product type are unobserved, some of this rate variation may reflect shifts in deposit composition rather than changes in posted or negotiated prices. These pricing dynamics corroborate the deposit-flow evidence: weak banks paid higher funding costs when large deposits were leaving under limited insurance, and the remaining rate premium under full coverage helped them attract back large balances.

5 Deposit Insurance and Credit Allocation

We now examine whether the funding relief from unlimited insurance affected bank lending. The insurance expansion reverses outflows of large previously uninsured deposits from weak banks; we ask whether these banks used the relief to sustain credit supply, and whether the additional credit went to riskier borrowers (Hsieh and Klenow, 2009; Acharya et al., 2022).

We begin by examining credit growth at the firm-bank level and ask whether lending patterns differ across banks with different pre-crisis loan-to-deposit ratios. To isolate changes in credit supply from changes in credit demand, we follow Khwaja and Mian (2008) and compare lending by different banks to the same firm within the same year. To capture different dimensions of firm risk, we analyze two risk measures: an ex-ante firm risk index combining leverage, profitability, and earnings volatility (defined in Section 2.4) and an ex-post default indicator equal to one for firms that default between 2009 and 2015.⁸ We estimate the following OLS regressions separately in 2008 and 2009:

$$\Delta \log(\text{Credit})_{bft} = \alpha_f + \alpha_b + \beta_t \text{Loan/Deposits}_{b,2007} \times \text{Firm Risk}_{f,t} + \epsilon_{bft}, \quad (5)$$

where $\Delta \log(\text{Credit})_{bft}$ denotes the annual change in log credit from bank b to firm f in year t . The variable $\text{Loan/Deposits}_{b,2007}$ is bank b 's de-meaned loan-to-deposit ratio at the end of 2007. The variable $\text{Firm Risk}_{f,t}$ is either the lagged ex-ante risk index or the ex-post default indicator. Firm fixed effects α_f isolate within-firm changes in credit amounts. Bank fixed effects α_b absorb bank-level differences in credit growth, including the level effect of the loan-to-deposit ratio. Standard errors are double-clustered at the bank and firm level.

Table 4 reports the credit-growth estimates. Before turning to borrower-risk interactions, columns 1–2 estimate a simpler firm fixed-effects specification with only banks' de-meaned 2007 loan-to-deposit ratio on the right-hand side. The negative coefficients imply that banks with higher loan-to-deposit ratios supply less credit to the same firm than other banks in both years. Columns 3–6 then estimate equation (5), adding bank

⁸Appendix Table A.7 confirms that defaulting firms are significantly weaker across observable characteristics.

fixed effects and interacting the loan-to-deposit ratio with firm risk, so the interaction coefficient tests whether banks with higher loan-to-deposit ratios supply more credit to riskier borrowers than banks with lower ratios do, conditional on firm and bank fixed effects. Using the ex-ante risk index, this interaction is negative and insignificant in 2008 and turns positive and significant in 2009; the ex-post default measure shows the same pattern. Since 2009 is the first full year of unlimited coverage, the reform-relevant test is whether this differential lending to riskier firms by banks with higher loan-to-deposit ratios increases from 2008 to 2009. The reported 2009-minus-2008 difference is positive and statistically significant for both risk measures. The benchmark evidence thus suggests that banks with higher loan-to-deposit ratios become more likely than other banks to direct credit growth toward riskier firms from 2008 to 2009.

A key challenge in interpreting these credit patterns is that weaker banks were exposed to many crisis-period shocks that could affect lending independently of the changes in deposit insurance. We address this concern by combining the loan-to-deposit ratio with banks' pre-crisis uninsured-deposit share. The loan-to-deposit ratio captures weak pre-crisis fundamentals, while the uninsured-deposit share captures banks' exposure to the removal of the insurance limit. We refer to banks with both a higher loan-to-deposit ratio and a high uninsured-deposit share as exposed banks.

Table 5 incorporates this exposure through the high uninsured-deposit-share indicator, equal to one for banks whose 2007 uninsured-deposit share is at or above the sample mean. Columns 1–2 first estimate a double-interaction specification, interacting *Loan/Deposits* with *High Uninsured*, which asks whether credit growth changes differentially at banks that are both weak and highly exposed to the insurance expansion. Columns 3–6 then add borrower risk by interacting this exposure term with *Firm Risk_{f,t}*. The coefficient of interest in these columns is therefore on the triple interaction of *Loan/Deposits*, *High Uninsured* and *Firm Risk*. Columns 1–2 include firm fixed effects, while columns 3–6 include firm and bank fixed effects as in equation (5); all specifications double-cluster standard errors at the bank and firm level and include lower-order terms where separately identified.

Columns 1–2 in Table 5 show that banks with both higher loan-to-deposit ratios and high uninsured-deposit shares contract credit under limited insurance and ex-

pand credit under unlimited insurance. The double interaction is negative in 2008 and positive in 2009. A one-standard-deviation increase in the loan-to-deposit ratio implies about 14 log points lower credit growth at high-uninsured banks in 2008, but about 17 log points higher credit growth in 2009. The 2009-minus-2008 difference is 31 log points and is statistically significant at the 1% level. This pattern matches the deposit-side evidence that the reform most affected weak banks with large previously uninsured deposit balances.

Columns 3–6 show which borrowers account for this relative expansion. The ex-ante firm risk index enters in columns 3–4, ex-post default in columns 5–6. In 2009, the triple interactions are positive for both borrower-risk measures, indicating that banks with both higher loan-to-deposit ratios and high uninsured-deposit shares increase credit disproportionately to riskier firms. Relative to 2008, the triple interaction in 2009 rises for both risk measures. Panel B of Table 5 plots the three interaction terms alongside their 2009-minus-2008 differences. The unconditional shares move the same way: at exposed banks, the share of gross credit expansion going to firms in that year’s top risk quartile rose from 7% in 2008 to 18% in 2009, while at other banks it fell from 12% to 7%. Thus, after coverage becomes unlimited, exposed banks direct more credit toward riskier firms.

Loan Rates and Borrower Risk. Appendix Tables A.4 and A.5 provide the loan-rate counterparts to the credit-growth results. Banks with higher loan-to-deposit ratios charge somewhat higher average loan rates, but the interactions with the firm risk index and defaulting-firm indicator are not statistically significant. In the uninsured-share specification, the loan-rate interactions are also small and statistically insignificant, as Panel B of Appendix Table A.5 shows. The credit reallocation toward riskier borrowers is therefore not matched by higher measured loan rates. This pattern suggests that the additional risk taken by weaker banks is not reflected in observed loan pricing.

Taken together, the lending results show that unlimited deposit insurance shifts weaker banks with greater exposure to the reform from relative credit contraction under limited insurance to relative credit expansion under unlimited insurance. This expansion is directed toward ex-ante riskier firms and firms with higher ex-post de-

fault rates, without a corresponding increase in measured loan rates.

6 Deposit Reallocation and Future Bank Failures

The deposit and lending evidence shows that unlimited insurance redirected funding toward banks with weak pre-crisis fundamentals and changed their credit supply. We now use bank failures during 2009–2015 as an ex-post benchmark for this weak-bank interpretation. The 2007 loan-to-deposit ratio structures the deposit-flow and lending results presented above. We test whether this ratio, together with realized deposit flows during the crisis, predicts subsequent bank failure.⁹

Table 6 reports cross-sectional bank-level logit regressions where the dependent variable equals one for banks that failed at any point between 2009 and 2015. The table also reports the area under the receiver operating characteristic curve (AUC), which measures how well a model ranks failed banks above surviving banks by predicted failure probability. An AUC of 0.5 corresponds to no predictive discrimination, while an AUC of 1 corresponds to perfect discrimination, so the statistic is useful for comparing the predictive content of parsimonious failure models.

We start with benchmark predictors of bank failure risk, all measured in 2007. The Tier 1 capital ratio is not statistically significant, and the AUC is 0.74 when the capital ratio enters alone. Adding bank size leaves the capital-ratio coefficient insignificant and lowers the AUC to 0.71. In contrast, the loan-to-deposit ratio performs markedly better: it is a positive and highly significant predictor of future bank failure, and achieves an AUC of 0.81; these effects are robust to additionally controlling for bank capitalization and size. These results support our use of the loan-to-deposit ratio as an ex-ante measure of bank weakness.

Next, we examine the relationship between bank-level deposit flows during the crisis and future bank failure. During 2008:Q3, when deposit insurance was still limited, lower deposit growth predicts subsequent bank failure: banks experiencing larger outflows are more likely to fail later. In 2008:Q4, after unlimited coverage was introduced,

⁹Appendix Table A.8 shows that banks failing between 2009 and 2015 already had weaker pre-crisis fundamentals, including higher loan-to-deposit ratios, faster boom-period credit growth, greater real-estate exposure, and lower median Tier 1 capital.

the relationship reverses. Higher deposit growth predicts subsequent failure, and is a better predictor of future bank failures than deposit outflows in 2008:Q3, as measured by the AUC. This reversal is striking. Under limited insurance, outflows predict later failure; under unlimited insurance, inflows predict later failure. The raw failure rates show the same pattern: 43% of banks in the top quartile of realized 2008:Q4 deposit growth failed between 2009 and 2015, against 17% of banks in the bottom three quartiles.

Columns 7–8 repeat the exercise with predicted deposit growth: the loan-to-deposit component of a regression of quarterly deposit growth on banks' 2007 loan-to-deposit ratio, capital ratio and size, with headquarters-region fixed effects absorbing local economic conditions. The other terms enter only as controls, so the loan-to-deposit coefficient holds them fixed while the predicted series varies with bank weakness alone. The reversal survives: predicted outflows in 2008:Q3 and predicted inflows in 2008:Q4 both signal later failure. Compared to the predictions using realized deposit growth in columns 5–6, isolating the weakness-driven component does sharpen the prediction: the AUC rises from 0.70 to 0.81 in 2008:Q3 and from 0.76 to 0.82 in 2008:Q4. Sorting banks on predicted 2008:Q4 deposit growth widens the gap in failure rates: 48% of the top quartile failed between 2009 and 2015, against 15% of the bottom three quartiles. The same fundamentals that drove deposits out of weak banks under limited insurance drew them back once coverage became unlimited. Because quarterly deposit growth is not reported for all banks in all quarters, Appendix Table [A.6](#) repeats the analysis on banks with consistently reported data and confirms the reversal.

Finally, a higher share of uninsured deposits at the bank level has more limited standalone predictive power. It is positive and marginally significant without controls, but loses precision once capital and size are included and turns negative once the loan-to-deposit ratio is added. This pattern is consistent with our interpretation of the uninsured share as exposure to the insurance reform rather than a standalone measure of bank weakness. Together, the results show that increasing deposit insurance stabilized weak banks' funding in the short run partly by reallocating deposits toward banks that later failed. This final evidence supports the paper's central interpretation: unlimited coverage prevented weak banks from further losing large previously

uninsured deposits, enabled continued credit supply, and reallocated deposits toward banks that were more likely to fail.

7 Conclusion

Deposit insurance changes are mostly made within financial crises; however, most of the empirical literature focuses on the effects of deposit insurance during normal times. In this paper, we study how increasing deposit insurance coverage during a crisis shapes deposit behavior and associated bank credit supply, including excessive bank risk-taking. For identification, we exploit a reform of Denmark’s deposit insurance scheme during the Global Financial Crisis in conjunction with comprehensive administrative datasets, including the universe of deposits—deposit register—and the credit register.

Increased deposit insurance triggers deposit reallocation from stronger to weaker banks. Under limited deposit insurance, uninsured deposit withdrawals are concentrated at banks with weaker ex-ante fundamentals, and expanded insurance reverses this targeted depositor behavior. Effects are stronger for richer individuals and for firms with higher equity. Despite that both retail and corporate depositors withdraw large (previously uninsured) deposits from weak banks under limited insurance, only retail depositors put deposits back into weaker banks under unlimited insurance. Under limited insurance, moreover, banks with weaker fundamentals increase deposit rates, and these higher rates decrease gradually over time with unlimited insurance. Furthermore, we find that exposed banks contract credit supply in the crisis under limited insurance but expand credit supply under unlimited insurance. However, this credit supply expansion by exposed banks with unlimited insurance disproportionately benefits very risky firms, measured either by ex-ante risk or by ex-post defaults. Finally, and consistent with previous results, banks that gained the most deposits under unlimited insurance are themselves more likely to fail ex-post, particularly those banks with weaker ex-ante fundamentals. These results suggest that increasing deposit insurance coverage can prolong the survival of fundamentally unsound banks and firms.

These findings contribute to our understanding of optimal deposit insurance de-

sign, particularly regarding appropriate coverage limits ([Dávila and Goldstein, 2023](#)). By demonstrating how coverage limits affect depositor behavior and the subsequent allocation of credit to lower-quality firms, our work informs the ongoing policy debate on deposit insurance design. Weak, highly exposed banks expand credit by roughly 31 log points more in 2009 than in 2008 relative to less exposed banks, easing funding-driven credit contraction, but this additional credit was disproportionately allocated to riskier firms, and the same weak banks are themselves substantially more likely to fail ex-post (an ex-ante loan-to-deposit ratio alone achieves an AUC of 0.81 for predicting failure). Policymakers must carefully calibrate coverage limits to balance financial stability with maintaining effective market discipline.

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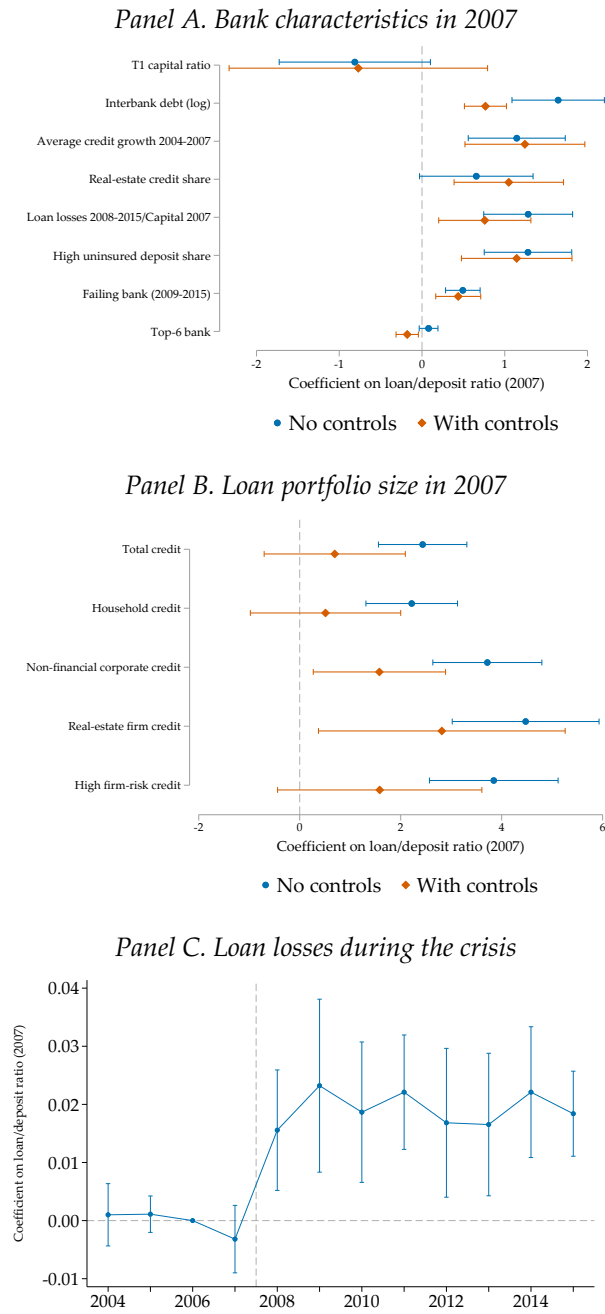
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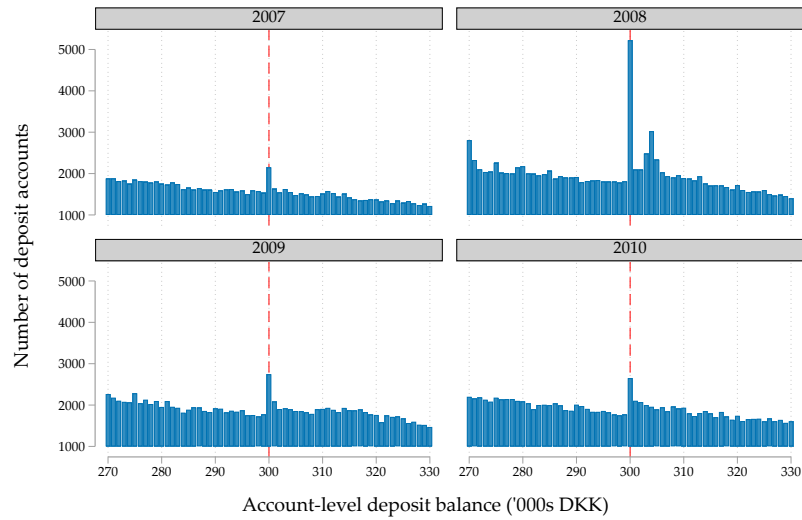
Figure 1: Bank Weakness and the Loan-to-Deposit Ratio



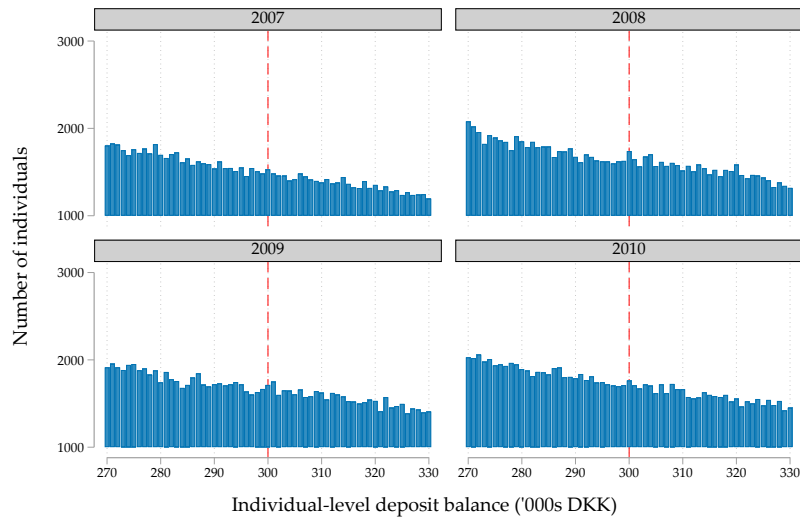
Notes: Panels A and B present cross-sectional bank-level regressions of 2007 bank characteristics on the 2007 loan-to-deposit ratio. Blue circles report specifications with only the loan-to-deposit ratio on the right-hand side; orange diamonds add 2007 bank controls for log assets, the Tier 1 capital ratio, and an indicator for the six largest banks, excluding a control when it is the dependent variable. Panel A uses bank-level characteristics as outcomes; continuous outcomes are standardized and indicator outcomes are left in levels; see Section 3.1 for the definitions of Panel A's outcome variables. Panel B uses log 2007 loan-portfolio amounts as outcomes. Panel C presents year-specific coefficients from bank-year regressions of loan losses scaled by total assets on the 2007 loan-to-deposit ratio interacted with year indicators; Panel C includes bank and year fixed effects and additionally interacts year indicators with 2007 controls for the de-meaned Tier 1 capital ratio, de-meaned log assets, and an indicator for the six largest banks; coefficients are normalized relative to 2006. Confidence intervals in Panels A and B use robust standard errors; Panel C clusters standard errors at the bank level.

Figure 2: Retail Deposit Bunching at the Insurance Limit

Panel A. Depositor-bank-level



Panel B. Individual-level



Notes: This figure plots the empirical distribution of retail deposit balances around the DKK 300,000 insurance threshold. Panel A reports depositor-bank-level balances; Panel B reports individual-level balances summed across all banks. Each point reports the number of balances in a DKK 1,000-wide range. The sample covers retail deposits in 2007–2010. The vertical line marks the 2007–2008 insurance limit; coverage was unlimited from October 2008 through September 2010. Source: Danish deposit register.

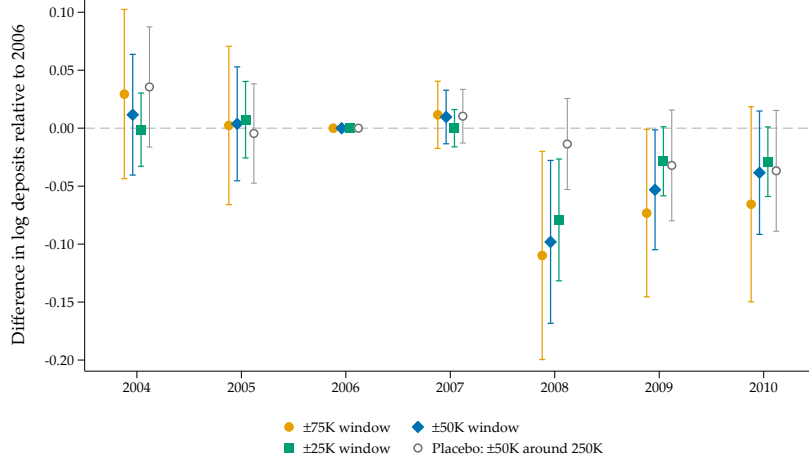
Table 1: Funding Stress Around the Deposit Insurance Reform

	2004–07	2008q1	2008q2	2008q3	2008q4	2008q3	2008q4	2008q4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: $\Delta \log$ total deposits								
Loan/deposit ratio	0.02 (0.01)	-0.01 (0.04)	-0.02 (0.03)	-0.07** (0.03)	0.16*** (0.04)	-0.02*** (0.00)	0.02* (0.01)	0.01** (0.01)
Observations	1,037	91	81	78	86	78	76	86
R^2	0.01	0.18	0.06	0.19	0.34	0.18	0.15	0.16
L/D measured in	2007q4	2007q4	2007q4	2007q4	2007q4	2008q2	2008q2	2008q3
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Δ deposit rate								
Loan/deposit ratio	-0.01 (0.06)	0.05 (0.18)	0.11 (0.14)	0.27** (0.13)	-0.82*** (0.18)	0.04** (0.02)	-0.17*** (0.04)	-0.11*** (0.03)
Observations	734	81	71	70	73	70	73	73
R^2	0.00	0.02	0.03	0.16	0.28	0.08	0.18	0.16
L/D measured in	2007q4	2007q4	2007q4	2007q4	2007q4	2008q2	2008q2	2008q3
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

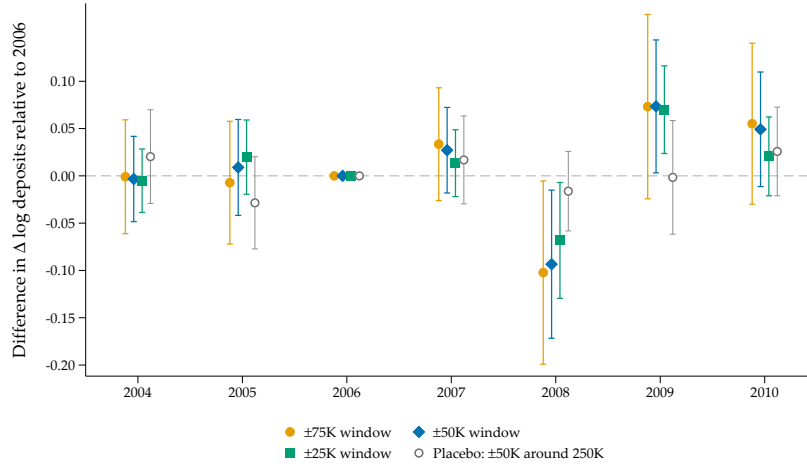
Notes: This table reports OLS estimates from cross-sectional regressions, across Danish banks, of the quarter-on-quarter change in log total deposits (Panel A) and in the deposit rate, in percentage points (Panel B), on the bank's loan-to-deposit ratio. Column headers give the quarter in which the regression is run; the row "L/D measured in" gives the quarter at which the loan-to-deposit ratio is measured. Observation counts vary because not every bank reports every quarterly variable; Appendix Table A.2 repeats the analysis on a balanced sample. Robust standard errors are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Figure 3: Deposit Growth Around the Insurance Limit

Panel A. Log deposit levels



Panel B. Log deposit changes



Notes: This figure plots event-study coefficients and 95% confidence intervals from regressions at the bank $\times$ deposit-bin $\times$ year level. Panel A's dependent variable is log deposits aggregated into DKK 25,000 deposit bins; Panel B's dependent variable is the annual change in that same log deposit measure. Deposit bins are constructed as described in Section 4.1, and include both retail and non-financial corporate deposits. Coefficients are for $\eta_t \times \text{Above } 300K_{kt} \times \text{Loan/Deposits}_{b,2007}$, where η_t denotes year indicators, $\text{Above } 300K_{kt}$ equals one for deposit bins above the DKK 300,000 deposit insurance threshold in year t , and $\text{Loan/Deposits}_{b,2007}$ is the 2007 loan-to-deposit ratio. Orange circles, blue diamonds, and green squares use windows of DKK 75,000, DKK 50,000, and DKK 25,000 around the DKK 300,000 threshold, corresponding to three, two, and one bins on each side; gray hollow triangles use the DKK 50,000 placebo window around DKK 250,000. Specifications include bank-year and bin-year fixed effects; standard errors are clustered at the bank level. Regression estimates appear in Table 2.

Table 2: Deposit Growth Around the Insurance Limit: Regression Estimates

	Log deposits				Δ Log deposits			
	300K threshold			250K placebo	300K threshold			250K placebo
	(1) [225K,375K]	(2) [250K,350K]	(3) [275K,325K]	(4) [200K,300K]	(5) [225K,375K]	(6) [250K,350K]	(7) [275K,325K]	(8) [200K,300K]
2004 $\times$ Above threshold $\times$ Loan/deposit ratio	0.03 (0.04)	0.01 (0.03)	-0.00 (0.02)	0.04 (0.03)	-0.00 (0.03)	-0.00 (0.02)	-0.01 (0.02)	0.02 (0.02)
2005 $\times$ Above threshold $\times$ Loan/deposit ratio	0.00 (0.03)	0.00 (0.02)	0.01 (0.02)	-0.00 (0.02)	-0.01 (0.03)	0.01 (0.03)	0.02 (0.02)	-0.03 (0.02)
2007 $\times$ Above threshold $\times$ Loan/deposit ratio	0.01 (0.01)	0.01 (0.01)	-0.00 (0.01)	0.01 (0.01)	0.03 (0.03)	0.03 (0.02)	0.01 (0.02)	0.02 (0.02)
2008 $\times$ Above threshold $\times$ Loan/deposit ratio	-0.11** (0.05)	-0.10*** (0.04)	-0.08*** (0.03)	-0.01 (0.02)	-0.10** (0.05)	-0.09** (0.04)	-0.07** (0.03)	-0.02 (0.02)
2009 $\times$ Above threshold $\times$ Loan/deposit ratio	-0.07** (0.04)	-0.05** (0.03)	-0.03* (0.01)	-0.03 (0.02)	0.07 (0.05)	0.07** (0.04)	0.07*** (0.02)	-0.00 (0.03)
2010 $\times$ Above threshold $\times$ Loan/deposit ratio	-0.07 (0.04)	-0.04 (0.03)	-0.03* (0.02)	-0.04 (0.03)	0.06 (0.04)	0.05 (0.03)	0.02 (0.02)	0.03 (0.02)
Above threshold $\times$ Loan/deposit ratio	0.09** (0.04)	0.06** (0.03)	0.03* (0.01)	0.06* (0.03)	-0.02 (0.03)	-0.02 (0.02)	-0.01 (0.02)	-0.01 (0.02)
Observations	3,528	2,352	1,176	2,352	3,450	2,300	1,150	2,300
R^2	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
Bank-year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Range-year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the regression estimates underlying Figure 3. Deposit bins of DKK 25,000 are constructed as described in Section 4.1. Columns 1–4 report the estimates for Figure 3, Panel A, where the dependent variable is log retail-plus-corporate deposits; columns 5–8 report the estimates for Figure 3, Panel B, where the dependent variable is the annual change in that same log deposit measure. The spanning header gives the dependent variable, and the second header row gives the deposit-balance window in DKK thousands. Within each dependent variable, the first three columns use windows of DKK 75,000, DKK 50,000, and DKK 25,000 around the DKK 300,000 threshold, corresponding to three, two, and one bins on each side; the fourth uses the DKK 50,000 placebo window around DKK 250,000. Reported coefficients are for $\text{Above } 300K_{kt} \times \text{Loan/Deposits}_{b,2007}$ and year interactions with that term, normalized relative to 2006; the first (unlabeled) row reports the 2006-level coefficient itself, and the year-labeled rows below report differences from this 2006 baseline. Specifications include bank-year and bin-year fixed effects; standard errors are clustered at the bank level and reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 3: Depositor-Bank Deposit Growth and Bank Weakness*Panel A. Baseline double-interaction with household and firm deposits*

	Individuals		Firms		Individuals + Firms	
	2008 (1)	2009 (2)	2008 (3)	2009 (4)	2008 (5)	2009 (6)
Loan/deposit ratio $\times$ Above DKK 300K	-0.12*** (0.04)	0.05** (0.02)	-0.25* (0.13)	0.00 (0.13)	-0.14*** (0.04)	0.04* (0.02)
Above DKK 300K	-0.36*** (0.02)	-0.19*** (0.01)	-0.51*** (0.03)	-0.43*** (0.03)	-0.36*** (0.02)	-0.20*** (0.01)
Observations	1,828,479	1,471,244	17,834	12,846	1,846,313	1,484,092
R^2	0.48	0.47	0.48	0.48	0.48	0.47
Depositor FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates from depositor-bank-year regressions of annual changes in log deposits for depositors with multiple bank relationships. The regressions compare deposit growth for depositor-bank relationships with prior-year balances above versus below the DKK 300,000 insurance limit. *Above DKK 300K*_{*t*-1} equals one if depositor *i*'s deposits at bank *b* exceeded DKK 300,000 in the previous year. Panel A estimates the interaction between this indicator and the de-meaned 2007 loan-to-deposit ratio separately for individuals, non-financial corporate depositors, and the pooled sample in 2008 and 2009. The estimation sample is restricted to depositors with multiple bank relationships in the estimation year; for 2009, depositors must also belong to the 2008 estimation sample. Specifications include depositor and bank fixed effects. Standard errors are double-clustered by depositor and bank and reported in parentheses. Panel B, on the following page, adds triple interactions with heterogeneity variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

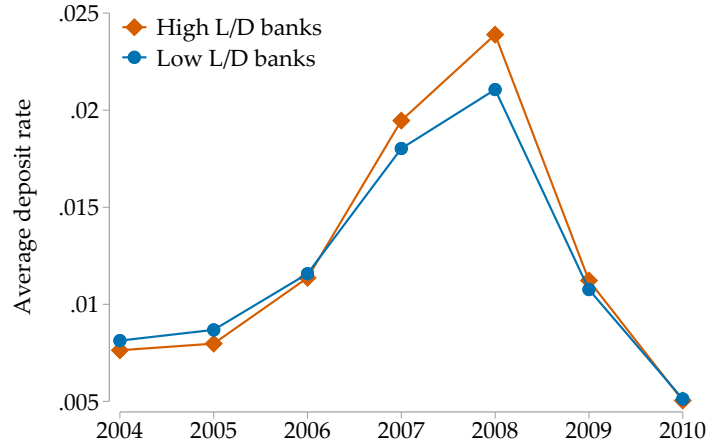
Table 3: Depositor-Bank Deposit Growth and Bank Weakness (continued)*Panel B. Triple interactions with heterogeneity variables*

	X = Uninsured share (2007)						X = Wealth		X = Equity ratio (2007)	
	Individuals		Firms		Individuals + Firms		Individuals		Firms	
	2008 (1)	2009 (2)	2008 (3)	2009 (4)	2008 (5)	2009 (6)	2008 (7)	2009 (8)	2008 (9)	2009 (10)
Loan/deposit ratio $\times$ Above DKK 300K $\times$ X	-0.14** (0.07)	0.08** (0.04)	-0.51 (0.31)	0.37 (0.32)	-0.17** (0.07)	0.08* (0.04)	-0.00* (0.00)	0.01** (0.00)	-1.32*** (0.46)	0.18 (0.57)
Loan/deposit ratio $\times$ Above DKK 300K	-0.00 (0.05)	0.01 (0.04)	0.16 (0.28)	-0.23 (0.31)	-0.00 (0.05)	0.00 (0.04)	-0.12*** (0.04)	0.05** (0.02)	0.30 (0.24)	-0.09 (0.28)
X $\times$ Above DKK 300K	-0.00 (0.02)	-0.04*** (0.01)	0.09 (0.08)	-0.22** (0.08)	0.00 (0.02)	-0.05*** (0.01)	-0.00 (0.00)	-0.01*** (0.00)	0.41*** (0.11)	0.12 (0.15)
Above DKK 300K	-0.35*** (0.01)	-0.16*** (0.01)	-0.57*** (0.06)	-0.26*** (0.08)	-0.36*** (0.01)	-0.16*** (0.01)	-0.35*** (0.02)	-0.19*** (0.01)	-0.65*** (0.06)	-0.44*** (0.07)
Observations	1,828,479	1,471,244	17,834	12,846	1,846,313	1,484,092	1,820,007	1,464,244	12,967	9,500
R^2	0.48	0.47	0.48	0.48	0.48	0.47	0.48	0.47	0.48	0.47
Depositor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

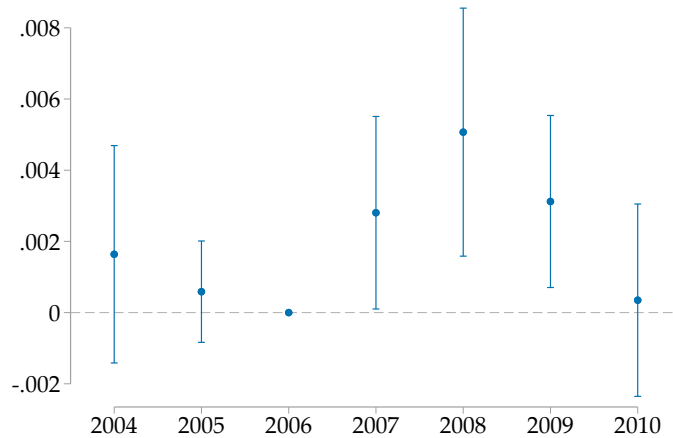
Notes: This panel continues Table 3. It reports OLS estimates from depositor-bank-year regressions of annual changes in log deposits for depositors with multiple bank relationships, adding a triple interaction between $Above\ DKK\ 300K_{t-1}$, the de-meaned 2007 loan-to-deposit ratio, and the heterogeneity variable X . $Above\ DKK\ 300K_{t-1}$ equals one if depositor i 's deposits at bank b exceeded DKK 300,000 in the previous year. X is the bank's 2007 uninsured-deposit share in columns 1–6, retail depositors' wealth in columns 7–8, and firms' 2007 equity ratio in columns 9–10. Wealth is measured in 2007 and standardized to mean zero and unit standard deviation. As in Panel A, the estimation sample is restricted to depositors with multiple bank relationships in the estimation year, and for 2009 depositors must also belong to the 2008 estimation sample. Specifications include depositor and bank fixed effects. Standard errors are double-clustered by depositor and bank and reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Figure 4: Deposit Rates Around the Insurance Expansion

Panel A. Average deposit rate (households + non-financial corporations)



Panel B. Dynamic regression: average deposit rate on loan/deposit ratio (2007)



Notes: This figure plots deposit-rate dynamics around the October 2008 expansion of deposit insurance. The bank-level deposit rate is the average of the deposit rates for individual and nonfinancial corporate depositors at the bank. Panel A plots the simple mean across banks in each year, separately for banks above and below the 2007 sample median loan-to-deposit ratio. Panel B reports coefficients and 95% confidence intervals from bank-year regressions of the same average deposit rate on year indicators interacted with banks' de-meaned 2007 loan-to-deposit ratio. The specification includes bank and year fixed effects and also interacts year indicators with de-meaned 2007 controls for the Tier 1 capital ratio and log assets, plus an indicator for the six largest banks. The plotted coefficients are normalized relative to 2006, and standard errors are clustered at the bank level.

Table 4: Firm-Bank Credit Growth and Borrower Risk

Dependent Variable: $\Delta \log \text{ credit}$	(1) 2008	(2) 2009	(3) 2008	(4) 2009	(5) 2008	(6) 2009
Loan/deposit ratio	-0.18*** (0.07)	-0.19*** (0.04)				
Loan/deposit ratio $\times$ Firm risk index			-0.32 (0.42)	0.73** (0.33)		
Loan/deposit ratio $\times$ Defaulting firm					-0.01 (0.08)	0.15* (0.09)
Observations	5,998	5,108	3,929	3,301	5,992	5,097
R^2	0.48	0.49	0.52	0.52	0.50	0.50
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE			Yes	Yes	Yes	Yes
Difference in coefficients, 2009 vs. 2008		-0.01 (0.08)		1.05** (0.53)		0.16* (0.12)

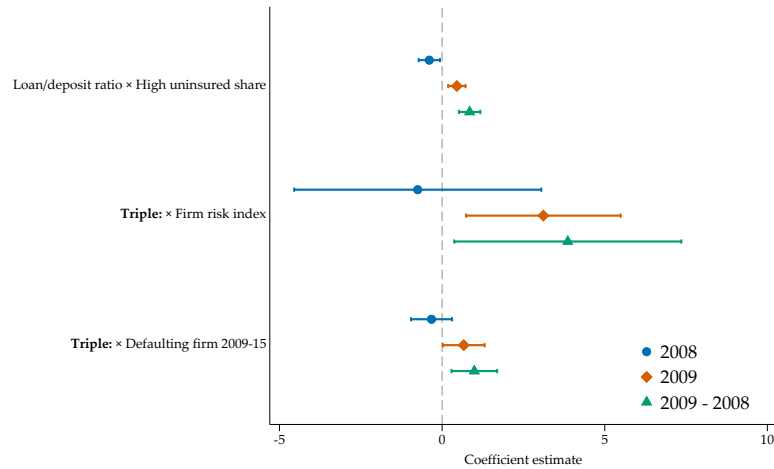
Notes: This table reports OLS estimates from firm-bank-year regressions of annual changes in log credit for non-financial firms with multiple bank relationships. Columns 1–2 report coefficients on banks’ de-meaned 2007 loan-to-deposit ratio. Columns 3–4 report its interaction with the lagged standardized firm risk index, and columns 5–6 report its interaction with an indicator for firms that default between 2009 and 2015. The firm risk index equals (Leverage – RoA) multiplied by RoA volatility, where RoA volatility is the standard deviation of RoA over the current and previous two years; the index is standardized by year and entered with a one-year lag. The row labeled “Difference in coefficients, 2009 vs. 2008” reports the 2009-minus-2008 difference in the displayed coefficient, with its standard error in parentheses. Loan-rate counterparts appear in Appendix Table A.4. Standard errors, double-clustered by bank and firm, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Stars on the difference row are based on a one-sided test, because the hypothesis is directional: unlimited coverage began in October 2008, so the alternative is $\beta_{2009} > \beta_{2008}$. Stars on all other rows are based on two-sided tests.

Table 5: Credit Growth, Bank Weakness, and Uninsured Deposit Exposure

Panel A. Regression estimates

Dependent Variable: $\Delta \log \text{ credit}$	(1) 2008	(2) 2009	(3) 2008	(4) 2009	(5) 2008	(6) 2009
High uninsured deposit share $\times$ Loan/deposit ratio	-0.39*	0.45***				
	(0.20)	(0.16)				
High uninsured deposit share	0.06*	-0.02				
	(0.03)	(0.03)				
Loan/deposit ratio	0.17	-0.62***				
	(0.19)	(0.15)				
Firm risk index $\times$ High uninsured deposit share $\times$ Loan/deposit ratio			-0.75	3.11**		
			(2.31)	(1.44)		
High uninsured deposit share $\times$ Firm risk index			0.27	-0.77***		
			(0.53)	(0.26)		
Loan/deposit ratio $\times$ Firm risk index			0.37	-2.14		
			(2.25)	(1.36)		
Defaulting firm $\times$ High uninsured deposit share $\times$ Loan/deposit ratio					-0.32	0.67*
					(0.38)	(0.39)
High uninsured deposit share $\times$ Defaulting firm					-0.02	-0.10
					(0.08)	(0.08)
Loan/deposit ratio $\times$ Defaulting firm					0.30	-0.43
					(0.37)	(0.38)
Observations	5,998	5,108	3,929	3,301	5,992	5,097
R^2	0.49	0.49	0.52	0.52	0.50	0.50
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE			Yes	Yes	Yes	Yes
Difference in interaction terms, 2009 vs. 2008		0.85***		3.86*		0.99**
		(0.26)		(2.72)		(0.55)

Panel B. Key interaction terms



Notes: Panel A reports OLS estimates from firm-bank-year regressions of annual changes in log credit for non-financial firms with multiple bank relationships. Columns 1–2 report the interaction between banks’ de-meanned 2007 loan-to-deposit ratio and an indicator for banks whose 2007 uninsured-deposit share is at or above the sample mean; columns 3–4 and 5–6 report the corresponding triple interactions with the lagged standardized firm risk index and with an indicator for firms defaulting between 2009 and 2015. The risk index is defined in the notes to Table 4. The row “Difference in interaction terms, 2009 vs. 2008” reports the 2009-minus-2008 difference in the displayed interaction coefficient, with its standard error in parentheses. Panel B plots the interaction coefficient from each pair of columns, top to bottom in the order 1–2, 3–4, 5–6: blue circles for 2008, orange diamonds for 2009, green triangles for the 2009-minus-2008 difference. Loan-rate counterparts appear in Appendix Table A.5. Standard errors are double-clustered by bank and firm. Stars mark significance at the 10% (*), 5% (**), and 1% (***) level and bars in Panel B are 90% intervals; both are two-sided except on the difference, which is tested one-sided against the directional alternative $\beta_{2009} > \beta_{2008}$, since unlimited coverage began in October 2008.

Table 6: Bank Failure, Weakness, and Crisis Deposit Growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Tier 1 capital ratio 2007	-0.07 (0.13)	-0.04 (0.10)		-0.03 (0.05)	-0.02 (0.05)	-0.09 (0.06)	-0.01 (0.04)	-0.05 (0.05)		-0.04 (0.08)	-0.03 (0.05)
Total assets 2007 (log)		0.18 (0.16)		-0.15 (0.17)	0.11 (0.13)	0.01 (0.15)	-0.14 (0.16)	-0.27 (0.19)		0.02 (0.16)	-0.21 (0.19)
Loan/deposit 2007			3.56*** (0.89)	3.87*** (1.02)							4.83*** (1.32)
Deposit growth 2008-q3					-7.44** (3.51)						
Deposit growth 2008-q4						3.83** (1.83)					
Predicted deposit growth 2008-q3							-1.38*** (0.40)				
Predicted deposit growth 2008-q4								1.62*** (0.43)			
High uninsured deposit share (>average)									0.91* (0.53)	0.86 (0.64)	-0.65 (0.82)
Constant	-0.42 (1.49)	-2.25 (2.27)	-5.83*** (1.24)	-4.50** (1.87)	-2.17 (1.43)	-0.65 (1.71)	-0.22 (1.69)	1.21 (1.93)	-1.77*** (0.38)	-1.37 (1.91)	-5.08** (2.27)
Observations	94	94	94	94	78	86	78	86	92	92	92
Pseudo R ²	0.02	0.04	0.20	0.21	0.08	0.10	0.19	0.23	0.03	0.05	0.23
AUC	0.74	0.71	0.81	0.82	0.70	0.76	0.81	0.82	0.61	0.70	0.84

Notes: This table reports logit estimates from bank-level regressions of bank failure between 2009 and 2015 on 2007 bank characteristics and 2008 deposit dynamics. The dependent variable equals one if the bank fails at any point between 2009 and 2015. Deposit growth is the quarterly change in log bank deposits. Predicted deposit growth is the loan-to-deposit component of bank-level regressions of the corresponding 2008 quarterly deposit-growth measure on banks' 2007 loan-to-deposit ratio, Tier 1 capital ratio and size, absorbing 2007 region fixed effects. It enters standardized, so columns 7 and 8 report the change in log-odds per standard deviation. High uninsured-deposit share is an indicator for banks whose 2007 uninsured-deposit share is at or above the sample mean. Bank controls are log 2007 assets and the 2007 Tier 1 capital ratio. AUC is the area under the receiver operating characteristic curve; here, it measures how well each specification ranks failed banks above surviving banks by predicted failure probability. The table uses the 2007 bank cross-section; observation counts vary with available quarterly deposit-growth measures. Balanced-sample results appear in Appendix Table A.6. Robust standard errors are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Appendix A Additional Figures and Tables

Table A.1: Pre-Crisis Summary Statistics

	Mean	SD	p10	p50	p90
Panel A. Banks					
Total assets (bn. DKK)	56.27	336.57	0.62	4.68	33.83
Number of employees	630.84	2,606.28	20.00	104.50	609.00
Deposits/Total assets	0.57	0.18	0.35	0.58	0.78
Loan/Deposit ratio	1.17	0.37	0.65	1.16	1.78
Tier 1 capital ratio (decimal)	0.13	0.09	0.08	0.12	0.19
Uninsured deposit share	0.42	0.15	0.26	0.39	0.54
Bank failures, 2009-2015	0.22	0.42	0.00	0.00	1.00
Panel B. Retail depositors					
Deposits (10,000 DKK)	10.52	110.73	0.09	1.94	22.70
Deposit rate	0.01	0.01	0.00	0.01	0.03
Deposit growth ($\Delta \log$)	0.06	0.67	-1.08	0.03	1.24
Gross wealth (100,000 DKK)	9.43	60.03	0.00	1.08	23.17
Has uninsured deposits	0.06	0.24	0.00	0.00	0.00
Has 2+ deposit banks	0.18	0.38	0.00	0.00	1.00
Panel C. Non-financial companies					
Deposits (100,000 DKK)	24.81	396.55	0.07	2.87	31.55
Deposit rate	0.02	0.02	0.00	0.01	0.05
Deposit growth ($\Delta \log$)	-0.07	1.09	-1.87	0.00	1.58
Has uninsured deposits	0.62	0.49	0.00	1.00	1.00
Has 2+ deposit banks	0.64	0.48	0.00	1.00	1.00
Credit (100,000 DKK)	64.28	677.40	0.18	6.52	80.62
Lending rate	0.08	0.05	0.02	0.07	0.13
Credit growth ($\Delta \log$)	0.04	0.65	-1.00	0.01	1.11
Has 2+ credit banks	0.48	0.50	0.00	0.00	1.00
Equity ratio	0.37	0.23	0.09	0.34	0.71
Firm risk index	0.05	0.17	0.00	0.02	0.11
Firm defaults, 2009-2015	0.26	0.44	0.00	0.00	1.00

Notes: This table reports summary statistics for banks, retail depositors, and nonfinancial firms as of December 2007. Panel A is at the bank level, Panel B is at the retail-depositor level, and Panel C is at the nonfinancial-firm level. Uninsured deposit share is deposits above DKK 300,000 divided by total deposits. Deposit and credit growth are log changes. The firm risk index equals (Leverage – RoA) multiplied by RoA volatility, where RoA volatility is the standard deviation of RoA over the current and previous two years. The table covers 94 banks, 5.3 million individuals, and 64,859 nonfinancial firms.

Table A.2: Funding Stress Around the Insurance Expansion: Balanced Sample

	RHS: 2007q4 L/D					RHS: 2008q2 L/D		RHS: 2008q3 L/D
	2004–07	2008q1	2008q2	2008q3	2008q4	2008q3	2008q4	2008q4
Panel A: $\Delta \log$ total deposits								
Loan/deposit ratio	0.01 (0.01)	-0.01 (0.04)	-0.03 (0.04)	-0.07** (0.04)	0.17*** (0.04)	-0.02*** (0.00)	0.02* (0.01)	0.01* (0.01)
Observations	789	69	69	69	69	69	69	69
R^2	0.01	0.24	0.06	0.19	0.33	0.18	0.15	0.14
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Δ deposit rate								
Loan/deposit ratio	-0.06 (0.07)	0.21** (0.08)	0.11 (0.14)	0.28** (0.14)	-0.79*** (0.17)	0.04** (0.02)	-0.16*** (0.04)	-0.12*** (0.03)
Observations	570	69	69	69	69	69	69	69
R^2	0.01	0.20	0.03	0.16	0.39	0.08	0.27	0.28
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the balanced-sample counterpart to Table 1: OLS estimates from cross-sectional bank-level regressions of the quarter-on-quarter change in log total deposits (Panel A) and in the deposit rate, in percentage points (Panel B), on the bank's loan-to-deposit ratio. The balanced sample comprises the 69 banks with non-missing deposit and rate observations in every quarter from 2007:Q1 through 2008:Q4. Column headers, the dates at which the ratio is measured, and the controls and fixed effects follow Table 1. Robust standard errors are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A.3: Cross-Bank Deposit Growth and Reform Exposure

	2004–2007		2008		2009	
	(1)	(2)	(3)	(4)	(5)	(6)
High uninsured deposit share $\times$ Loan/deposit ratio		-0.01 (0.02)		-0.06** (0.02)		-0.00 (0.03)
Loan/deposit ratio	-0.01 (0.01)	-0.01 (0.01)	-0.00 (0.01)	0.03* (0.02)	-0.04 (0.03)	-0.03 (0.02)
High uninsured deposit share		0.01* (0.01)		0.02 (0.01)		-0.01 (0.01)
Observations	21,733,511	21,733,511	5,659,247	5,623,065	5,634,952	5,598,566
R^2	0.00	0.00	0.00	0.00	0.00	0.00
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates from depositor-bank-year regressions pooling individual and nonfinancial corporate depositors. The dependent variable is the annual change in log deposits. Columns 1–2 pool 2004–2007; columns 3–4 use 2008; columns 5–6 use 2009. Odd-numbered columns include the de-meaned 2007 loan-to-deposit ratio and bank controls. Even-numbered columns add an indicator for banks whose 2007 uninsured-deposit share is at or above the sample mean and its interaction with the de-meaned 2007 loan-to-deposit ratio. Bank controls consist of the de-meaned 2007 Tier 1 capital ratio, de-meaned 2007 log assets, and an indicator for the six largest banks. Specifications do not absorb fixed effects. Standard errors, double-clustered by depositor and bank, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A.4: Loan Rates, Bank Weakness, and Borrower Risk

Dependent Variable: Loan rate	(1) 2008	(2) 2009	(3) 2008	(4) 2009	(5) 2008	(6) 2009
Loan/deposit ratio	0.01** (0.00)	0.01* (0.00)				
Loan/deposit ratio $\times$ Firm risk index			-0.02 (0.03)	0.02 (0.02)		
Loan/deposit ratio $\times$ Defaulting firm					-0.01 (0.01)	-0.00 (0.01)
Observations	8,852	7,596	5,702	4,864	8,850	7,592
R ²	0.50	0.53	0.51	0.55	0.51	0.55
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE			Yes	Yes	Yes	Yes
Difference in coefficients, 2009 vs. 2008		-0.00 (0.00)		0.03 (0.04)		0.01 (0.01)

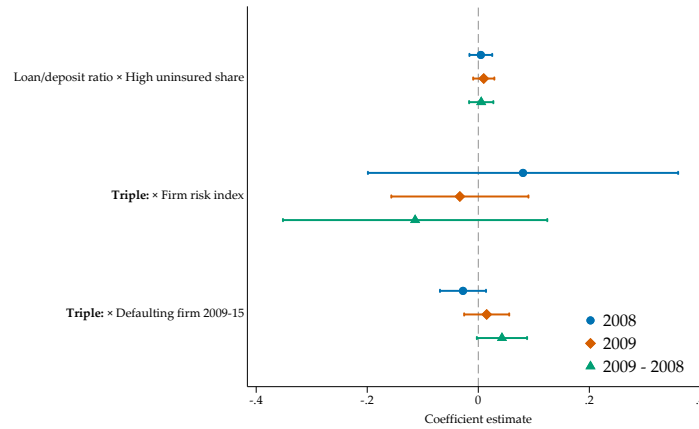
Notes: This table reports OLS estimates from firm-bank-year regressions of loan rates (expressed as decimals; see Section 2.2) for non-financial firms with multiple bank relationships. Columns 1–2 report coefficients on banks’ de-meaned 2007 loan-to-deposit ratio. Columns 3–4 report its interaction with the lagged standardized firm risk index, and columns 5–6 report its interaction with an indicator for firms that default between 2009 and 2015. The firm risk index equals (Leverage – RoA) multiplied by RoA volatility, where RoA volatility is the standard deviation of RoA over the current and previous two years; the index is standardized by year and entered with a one-year lag. The row labeled “Difference in coefficients, 2009 vs. 2008” reports the 2009-minus-2008 difference in the displayed coefficient, with its standard error in parentheses; none of these differences is statistically significant. This table is the loan-rate counterpart to Table 4. Standard errors, double-clustered by bank and firm, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A.5: Loan Rates, Bank Weakness, and Uninsured Deposits

Panel A. Regression estimates

Dependent Variable: Loan rate	(1) 2008	(2) 2009	(3) 2008	(4) 2009	(5) 2008	(6) 2009
High uninsured deposit share $\times$ Loan/deposit ratio	0.00 (0.01)	0.01 (0.01)				
High uninsured deposit share	-0.00 (0.00)	0.00 (0.00)				
Loan/deposit ratio	0.00 (0.01)	-0.00 (0.01)				
Firm risk index $\times$ High uninsured deposit share $\times$ Loan/deposit ratio			0.08 (0.17)	-0.03 (0.08)		
High uninsured deposit share $\times$ Firm risk index			-0.02 (0.03)	-0.00 (0.02)		
Loan/deposit ratio $\times$ Firm risk index			-0.09 (0.16)	0.05 (0.07)		
Defaulting firm $\times$ High uninsured deposit share $\times$ Loan/deposit ratio					-0.03 (0.03)	0.02 (0.02)
High uninsured deposit share $\times$ Defaulting firm					-0.01 (0.01)	-0.01 (0.01)
Loan/deposit ratio $\times$ Defaulting firm					0.02 (0.02)	-0.01 (0.02)
Observations	8,852	7,596	5,702	4,864	8,850	7,592
R^2	0.50	0.53	0.51	0.55	0.51	0.55
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE			Yes	Yes	Yes	Yes
Difference in interaction terms, 2009 vs. 2008		0.01 (0.02)		-0.11 (0.19)		0.04 (0.04)

Panel B. Key interaction terms



Notes: This table is the loan-rate counterpart to Table 5. Panel A reports OLS estimates from firm-bank-year regressions of loan rates (expressed as decimals; see Section 2.2) for non-financial firms with multiple bank relationships. Columns 1–2 report the interaction between banks’ de-meaned 2007 loan-to-deposit ratio and an indicator for banks whose 2007 uninsured-deposit share is at or above the sample mean; columns 3–4 and 5–6 report the corresponding triple interactions with the lagged standardized firm risk index and with an indicator for firms defaulting between 2009 and 2015. The risk index is defined in the notes to Table 4. The row “Difference in interaction terms, 2009 vs. 2008” reports the 2009-minus-2008 difference in the displayed interaction coefficient, with its standard error in parentheses; none of these differences is statistically significant. Panel B plots the interaction coefficient from each pair of columns, top to bottom in the order 1–2, 3–4, 5–6: blue circles for 2008, orange diamonds for 2009, green triangles for the 2009-minus-2008 difference. Standard errors are double-clustered by bank and firm. Stars mark significance at the 10% (*), 5% (**), and 1% (***) level and bars in Panel B are 90% intervals; both are two-sided except on the difference, which is tested one-sided against the directional alternative $\beta_{2009} > \beta_{2008}$, since unlimited coverage began in October 2008.

Table A.6: Bank Failure, Weakness, and Deposit Growth: Balanced Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Tier 1 capital ratio 2007	-0.03 (0.10)	-0.02 (0.08)		-0.02 (0.04)	-0.02 (0.05)	-0.09 (0.06)	-0.02 (0.04)	-0.02 (0.04)		-0.03 (0.07)	-0.01 (0.04)
Total assets 2007 (log)		0.08 (0.14)		-0.34* (0.20)	0.03 (0.12)	-0.10 (0.16)	-0.34* (0.20)	-0.34* (0.20)		0.01 (0.16)	-0.24 (0.23)
Loan/deposit 2007			4.07*** (1.19)	4.91*** (1.36)							6.70*** (1.96)
Deposit growth 2008-q3					-8.69** (3.97)						
Deposit growth 2008-q4						4.62** (1.94)					
Predicted deposit growth 2008-q3							-1.76*** (0.49)				
Predicted deposit growth 2008-q4								1.81*** (0.50)			
High uninsured deposit share (>average)									0.57 (0.57)	0.53 (0.74)	-1.68 (1.03)
Constant	-0.87 (1.16)	-1.70 (1.88)	-6.43*** (1.61)	-4.32** (2.07)	-1.70 (1.38)	0.08 (1.74)	1.41 (1.99)	1.36 (1.98)	-1.53*** (0.39)	-1.26 (1.65)	-6.92** (3.01)
Observations	74	74	74	74	74	74	74	74	74	74	74
Pseudo R^2	0.01	0.01	0.22	0.24	0.09	0.12	0.24	0.24	0.01	0.02	0.28
AUC	0.73	0.67	0.82	0.84	0.71	0.79	0.84	0.84	0.57	0.66	0.85

Notes: This table reports the balanced-sample counterpart to Table 6. The dependent variable equals one if the bank fails at any point between 2009 and 2015. Deposit growth is the quarterly change in log bank deposits. Predicted deposit growth is the loan-to-deposit component of bank-level regressions of the corresponding 2008 quarterly deposit-growth measure on banks' 2007 loan-to-deposit ratio, Tier 1 capital ratio and size, absorbing 2007 region fixed effects. It enters standardized, so columns 7 and 8 report the change in log-odds per standard deviation. High uninsured-deposit share is an indicator for banks whose 2007 uninsured-deposit share is at or above the sample mean. Bank controls are log 2007 assets and the 2007 Tier 1 capital ratio. AUC is the area under the receiver operating characteristic curve; here, it measures how well each specification ranks failed banks above surviving banks by predicted failure probability. The balanced sample comprises 74 banks observed throughout 2007:Q1–2008:Q4. Robust standard errors are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A.7: Summary Statistics for Defaulting and Surviving Firms

	Surviving firms			Defaulting firms ('09-'15)			Diff
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
Total assets (DKK million)	98.36	1,841.40	7.41	20.40	108.38	4.48	77.96**
Sales (DKK million)	98.36	1,841.40	7.41	20.40	108.38	4.48	77.96**
Employees	42.11	382.99	10.00	19.70	70.58	8.00	22.41***
Return on assets	0.04	0.19	0.05	-0.04	0.25	0.01	0.08***
Leverage	0.70	0.18	0.73	0.77	0.17	0.81	-0.07***
Equity/assets	0.30	0.18	0.27	0.23	0.17	0.19	0.07***
Firm risk index	0.05	0.11	0.02	0.07	0.11	0.04	-0.02***
Real-estate and construction firm	0.17	0.38	0.00	0.20	0.40	0.00	-0.03***

Notes: This table reports means, standard deviations, and medians for surviving and defaulting nonfinancial firms in the credit-register analysis. Defaulting firms are firms classified as defaults between 2009 and 2015, as described in Appendix A.2. The firm risk index equals (Leverage – RoA) multiplied by RoA volatility, where RoA volatility is the standard deviation of RoA over the current and previous two years. The difference column reports surviving-minus-defaulting mean differences. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A.8: Summary Statistics for Failed and Surviving Banks

	Surviving banks			Failed banks		
	Mean	Std. Dev.	p50	Mean	Std. Dev.	p50
Total assets (bn DKK)	87.72	434.65	4.63	14.87	17.35	7.82
Loan-to-deposit ratio	1.11	0.32	1.11	1.47	0.29	1.49
Tier 1 capital ratio (%)	12.67	4.83	11.80	11.39	9.68	9.10
Avg. credit growth 2004-2007	0.24	0.13	0.22	0.35	0.14	0.33
Interbank debt (log, std)	5.60	3.95	6.00	7.31	2.59	7.64
Real-estate credit share	0.35	0.22	0.29	0.46	0.17	0.46
Loan losses 2008-2015/Capital 2007	0.98	0.60	0.91	1.23	0.75	0.99
Top-6 bank	0.11	0.31	0.00	0.00	0.00	0.00

Notes: This table reports means, standard deviations, and medians for surviving and failed banks. Failed banks are banks classified as failures between 2009 and 2015, as described in [Appendix A.1](#).

A.1 Bank Failure Classification

Bank exits in our sample arise for two broad reasons: mergers and acquisitions (M&A) and financial distress leading to failure. The supervisory registers record exits but not their causes. We therefore classify distress-driven failures using observable deterioration in bank fundamentals.

The classification uses the exit year as the point at which fundamentals should reveal distress. Banks acquired or consolidated for reasons unrelated to solvency need not show this deterioration, so exit status alone would overstate failures. We classify an exit as a failure only when exit-year fundamentals signal distress.

We date each bank's exit at the bank-year level. A bank exits if the last year it appears in the panel falls within 2008–2015 and precedes the final sample year; in other words, the bank drops out of the panel rather than being right-censored. The exit year is the last observed year.

We assess distress with five exit-year indicators benchmarked against all banks in the same year. Because banks constitute a single industry, the benchmark requires no further sectoral adjustment. Three indicators flag lower-tail weakness, defined as values at or below the 20th percentile: the two-year-average growth rate of bank capital, the two-year-average growth rate of total assets, and the once-lagged solvency ratio (the regulatory capital-adequacy ratio).¹⁰ Two indicators flag upper-tail distress, defined as values at or above the 80th percentile: the accumulated impairment ratio and loan losses scaled by total assets. These measures capture low balance-sheet growth, weak capitalization, accumulated impairments, and current loan losses.

We classify a bank as a failure if at least three of the five indicators signal distress in the exit year. We also classify a bank as failing, regardless of the indicator count, if it reports negative core Tier 1 capital after deductions or zero employees in the exit year. These absolute triggers capture insolvency or a bank that has ceased operating.

Finally, we supplement the screen with hand-collected public default dates for the eight publicly traded Danish banks that failed during the global financial crisis. Any

¹⁰The two growth-rate indicators average each bank's annual growth over the exit year and the preceding year; the underlying annual growth rates are winsorized at the 5th and 95th percentiles before averaging.

bank with a recorded public default date is classified as a failure and assigned to the corresponding time window. This overlay captures listed-bank failures whose public failure date does not coincide with exit from the panel. We then separate banks failing in 2007–2008 from those failing in 2009–2015; the latter window defines the failed-bank indicator used throughout the paper. Appendix [A.2](#) applies the same logic to firm defaults.

A.2 Firm Default Classification

Firm exits arise for several reasons, including mergers and acquisitions (M&A), voluntary closures, restructurings, and financial distress leading to default. The administrative registers record exits but not their causes. We therefore classify distress-driven defaults using observable deterioration in firm fundamentals, mirroring the bank-failure procedure with two refinements suited to firms.

The first refinement is industry-year benchmarking. Firms differ more across sectors than banks do, so distress cutoffs should compare a firm with peers facing similar sectoral and macroeconomic conditions. The second refinement allows for incomplete accounting data: rather than requiring a fixed number of triggered indicators, we use the share of available indicators that signal distress.

We date each firm's exit from the business register. The primary exit year is the year its legal registration ends, based on the recorded legal end date. If the firm has no recorded legal end date, we use the last year it appears in the panel, provided that year precedes the end of the sample. Exit years after 2015 fall outside our window, so firms dated to exit after 2015 are right-censored and treated as surviving through the sample. For exits inside the window, we evaluate fundamentals in the exit year.

We assess distress with five indicators. Four are two-year averages over the exit year and the preceding year, which limits the influence of single-year fluctuations.¹¹ These four indicators are benchmarked within two-digit industry by year. The first flags low profitability: return on assets at or below the 20th percentile of the industry-year distribution. The second and third flag low sales growth and low employment growth, again using the bottom industry-year quintile. The fourth flags high leverage, defined as leverage at or above the 80th percentile of the industry-year distribution. The fifth indicator is absolute rather than relative: negative book equity, defined as equity at or below zero. Together, the indicators capture weak profitability, contracting activity, high leverage, and insolvency.

For each exiting firm, we compute the share of available indicators that signal dis-

¹¹We compute return on assets, sales growth, employment growth, and leverage as two-year averages over the exit year and the preceding year, and winsorize the underlying growth rates at the 5th and 95th percentiles before averaging.

stress: the number of triggered indicators divided by the number that are non-missing in the exit year. We classify an exiting firm as a default when at least half of its available indicators signal distress, and we code non-exiting firms as non-defaulting. We then separate firms defaulting in 2007–2008 from those defaulting in 2009–2015; the latter window defines the defaulting-firm indicator used throughout the credit analysis. Appendix Table A.7 shows that firms classified as defaulting are observably weaker than surviving firms before exit.